Reading Positivity from the Weights Alone

A general-purpose instrument in determinantal quantum Monte Carlo, scored against a criterion computed from the Hamiltonian

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Version 0.2.3. November 2026.

Abstract

The fermion sign problem in determinantal quantum Monte Carlo is normally diagnosed from the average sign, a quantity that costs `O(1/<sgn>^2)` to resolve and is identically 1 until the problem is already large. This paper asks a different question: **given only the weights a running simulation already holds – no Hamiltonian, no knowledge of the decoupling, and no constant chosen by the caller – what can be decided about its positivity?**

The answer is a build check a run can perform on itself. §4 derives an identity between the two spin channels' determinants: `ln|det_up| - ln|det_dn|` is an exact affine function of the field sum `sum(x)`. A normalised alignment saturates at 1 on an affine relation and cannot on a non-affine one, and it is invariant to both the offset and the slope, so the check needs neither the decoupling constant nor `tr(K)` – only two scalars every determinant QMC code already forms each sweep. On a lattice built with a periodic wrap where open was intended, the read fires at `beta = 1` to `3` while **not one negative weight has appeared**, every standard health check passes, and the criterion computed on the *intended* lattice still reports sign-free. It stands eleven orders above the correctly-built control and between `5.5` and `91` times its own seed spread. That is a realistic, common and expensive lattice-assembly bug, caught at a beta where a pilot run costs minutes.

For two scalar columns the read is the Pearson correlation of the centred columns, exactly, with one addition: it is standardised against an exact re-pairing null and returns `0` rather than a small correlation where the value does not clear it (§1.1). The cost is `O(n)` on columns already computed, against a diagnostic that otherwise costs `O(1/<sgn>^2)`.

A second reading classifies the run, from the weight cloud alone. `concentration` on the cloud's `(Re, Im)` frame returns a *directional* statistic `resultant` – the mean resultant length, which for a weight cloud is `|<w/|w|>|` – and an *axial* one `focus`, the leading eigenvalue of the cloud's orientation matrix. Because the rows are unit vectors in the plane, `focus` lies in `[ 1/2, 1]`, with `1/2` the isotropic floor and `1` exactly a rank-one cloud. The pair classifies: `focus = 1` with `resultant = 1` is sign-free; `focus = 1` with `resultant < 1` is a real sign problem, because rank one in the plane means the phase takes two values `pi` apart and a phase confined to `Z2` is what a sign *is*; `focus < 1` is a phase problem no rotation reaches. Measured, `focus` is `1.0000` on every real-weight row and `0.543` to `0.833` on every genuinely complex one.

The classification decides which estimator is safe. A global phase cancels in `<O> = sum(O w)/sum(w)`. The estimator carried over from real weights, `mean(Re w)/mean(|w|)`, reads `cos( theta)` – falling from `0.93500` to `0.25011` across rotations that leave `|<w>|` unmoved – and would overstate a cost that goes as `1/<sgn>^2` by fourteen times.

The reading is validated against an independent oracle computed from the Hamiltonian, and neither side is given the other's input. A configuration's weight is a product of two determinants, one per spin channel; on a bipartite lattice at half filling they change sign **in lockstep** – around 11% of configurations at `beta = 12`, on exactly the same ones – so the product's sign never changes. That lockstep follows from an identity between the two determinants, derived and verified to `1e-14` in two field distributions whose closed-form constants differ by 4% and each fail in the other by fourteen orders. The identity holds exactly when `K` admits a diagonal-unitary conjugation, by either of two routes – `S K S^-1 = -K`, which asks every cycle of `K`'s support graph to be even, or `S K S^-1 = -conj(K)` , which admits odd cycles carrying half-odd-integer flux. For real `K` the two coincide and reduce to a 2-colouring with a zero diagonal: `O(N^2)`, no eigenvalues, no determinants, no field.

That criterion is the standard one: for real `K` the two-colouring is bipartiteness and the zero diagonal is the absence of a staggered potential, the long-known particle-hole route to positivity. No new positivity criterion is claimed. It is the yardstick, and the agreement figures quoted in §5 – 69 one-body matrices with no mismatch, and the read agreeing with the criterion on 15 of 15 lattices from the output alone – are checks that this rig computes the known criterion correctly, which is what licenses the build check above. The oracle sees `K` and never a configuration; the read sees two logged columns and never `K`.

The read has one blind spot, and it is the same fact as its scale invariance: a wrong decoupling constant changes the slope, leaves the relation affine, and does not desaturate the read. The direct residual catches that case at `1.25`, so the two checks are complementary (§6).

The scope is measured, and it separates a comparison from a one-sided read. A phase common to both channels is invisible to a comparison between them: systems whose negative fractions differ by 33 percentage points give identical reads. The ceiling on what any such read can deliver is exact: correct importance sampling draws at `|w|`, so Kish's effective sample size of those weights is exactly `n <sgn>^2`, a property of the weights and therefore a ceiling on every aggregation of them. The read detects onset rather than severity: it moves on rows where `<sgn>` is still identically 1, and severity is set by a crossing count the read has already summed away.

A note on measurement. Every empirical quantity in this paper is a deterministic read of a configuration produced by the determinantal quantum Monte Carlo sampler in `research/`

code/. The sampler is gated against a brute-force enumeration of every auxiliary field at 5e-15. No fitted parameter is supplied to any read, and one constant is: the particle-hole zero 1/2 of §7's own-zero read, which follows from G_up[i,i] + G_dn[i,i] = 1 rather than being tuned. The one threshold in play is the library's own significance level far = 0.05, left at its default; it gates resolved, which §6.2 reports on the degenerate input. Elsewhere a coefficient that appears is derived, and the derivation is checked by requiring the wrong coefficient to fail.

Which results use the instrument, and which do not. Of the 54 experiments cited here, 28 read through the open-source *Entroptics* instrument (github.com/Agience/entroptics, version 0.2.10, doi:10.5281/zenodo.24063501) and 26 do not. All three checks the argument rests on – the lockstep, the identity, and the conditioned frame – are instrument-free. §§3-5 derive and verify the identity and its criterion from the one-body matrix; §7 reads the output with the instrument. Neither side is given the other's input – the read never receives K, the criterion never receives a configuration – and they agree on 15 of 15 lattices. Reads are reached through one domain adapter (research/code/entroptics_adapter.py) which names each library read after the question §7 asks of it and holds the version the figures were read through.

What is proved. The paper's algebraic statements are machine-checked in Lean 4 / Mathlib (§11, research/lean/), sorry-free and on the three foundational axioms: the log-determinant identity, the positivity criterion and its two routes, the closed door on doping, the effective-sample ceiling, magnitude blindness at every exponent, and what the alignment read guarantees. Every machine-precision figure quoted here is printed by the file it is attributed to.

1. What is normally measured

Determinantal quantum Monte Carlo [1] evaluates a fermionic partition function by decoupling the interaction into an auxiliary field and integrating the fermions out, leaving a sum over field configurations with weight w(x). When w is non-negative it is a probability density and the sum is a Monte Carlo average. When it is not, the standard workaround samples |w| and carries the sign as an observable:

```
<O> = <O sgn>_|w| / <sgn>_|w| ,        <sgn> = Z / Z_||
```

The denominator is the average sign, and it is exponentially small [4]: <sgn> = exp(-beta N df) with df the free-energy density difference between the physical ensemble and the one that samples |w|. Both numerator and denominator are then small numbers computed as differences of large ones, and the relative error of the ratio grows like 1 / sqrt(M <sgn>^2). The cost of a fixed accuracy is therefore exponential in beta N. Deciding the general case is NP-hard [5], so no representation-independ ent cure is available and the practical question is always which representation a given model admits.

That is the quantity this paper is about. The average sign is an average of a binary, so where the two channels are locked it returns 1.00000 +- 0.00000 on every configuration, and an average of a constant has no derivative: it carries no information about how close the system is to losing the lock.

That is a statement about one regime. Where <sgn> does vary it carries a great deal: Mondaini, Tarat and Scalettar [19] link it quantitatively to quantum critical behaviour, and read the low average sign of the doped square-lattice Hubbard model as reflecting the onset of pseudogap physics rather than only obstructing its simulation. This paper's reading is aimed at the complementary regime, where the sign is identically 1 and has nothing to move, and the order parameter of §7 is offered for the approach to that boundary.

1.1 The reads, as formulas

The instrument is general-purpose and its reads carry its own names. Every one used in this paper is a formula on the frame it is handed, and each reduces to a standard statistic or is stated here as one. A frame W is an n x k array – n draws, k features – and W~ is W with each column's mean removed. Reads are reached through research/code/entroptics_adapter.py, which names each after the question asked of it.

coupling(A, B), the two-channel read – §§3-7. A and B are frames sharing their n. The read is the normalised alignment of A~ and B~, and it returns:

- strength, in [-1, 1]. **For k = 1 this is exactly the Pearson correlation coefficient of the two centred columns**, agreeing with numpy.corrcoef to 1e-16 on affine, noisy-affine and weakly-related pairs. It is invariant to an offset and a positive scale on either column (§11, strength_affine_invariant), which is why §6 needs neither lambda nor tr(K); it is bounded by Cauchy-Schwarz (abs_strength_le_one); and it saturates at +-1 exactly when the two columns are affinely related (abs_strength_eq_one_iff).
- z and resolved, against the read's **exact re-pairing null**: the null re-pairs each frame row with every weight, giving the null distribution's mean and variance in closed form rather than asymptotically, so a sigma here is quoted with no constant supplied. Where the value does not clear its own null the read returns strength = 0 and resolved = False rather than a small correlation – on an independent pair of

columns `corrcoef` reads `0.086` and the read reads `0.0` at `z = 1.71`.

- `tightness`, the leading singular value's share of the squared cross-covariance spectrum, `s_1^2 / sum_i s_i^2`. It is `1` when the whole coupling sits in a single mode.

`concentration(P)`, the one-sided cloud read – §7.1. `P` is the `(Re, Im)` frame of the unit-modulus weights `u_i = w_i / |w_i|`, one row a draw. It returns a *directional* statistic and an *axial* one, and the pair is what classifies:

- `resultant = |mean(u)|`, the mean resultant length. On a weight cloud this **is** `|<w/|w|>|`, by construction rather than by measurement.
- `focus`, the leading eigenvalue of the orientation matrix `T = (1/n) sum_i u_i u_i^T`. Because the `u_i` are unit vectors in the plane, `tr(T) = 1` and therefore **`focus` lies in `[1/2, 1]`**: `1/2` is the isotropic floor, and `1` is exactly a rank-one cloud, every phase on one line through the origin. The two differ where it matters – an antipodal cloud reads `resultant ~ 0` and `focus = 1`.

  The floor is why `focus` is read against 1 and not against 0. §7.1's complex rows run `0.543` to `0.833`, which is a span from just above the isotropic floor to strongly axial; what every one of them shares is being below `1`, which is the definitional value the classification uses.

`carriage(...)`, the evidence read – §7, §9. `effective_n` is Kish's effective sample size `(sum w)^2 / sum w^2`. On a `+-1` sign sequence that is exactly `n <sgn>^2` (§11, `kish_of_signs`). `carried` is the same read's statement of how much of a frame a set of weights accounts for, standardised against the same exact re-pairing null; `1` is the value when the weights carry nothing, which is how §8's table is read.

`Screen().balance`, the own-zero read – §7. Defined where it is used: the joint residual `||r||^2` against the exact no-drift moments `E||r||^2 = tr(S)/T` and `Var||r||^2 = 2 tr(S^2)/T^2`, which fix a scaled chi-square whose degrees of freedom are the participation ratio `(tr S)^2 / tr(S^2)`.

`spectral_optics`, the single-channel read – §9.2b, §10. `top_share` is the leading eigenvalue's share of the covariance spectrum, `lambda_1 / sum_i lambda_i`; `attenuation`, `dominance` and `phase` are further summaries of the same spectrum, and `resolved_modes` is a count against a floor derived from the data. §9.2b uses all of them only through a **rank** correlation, which depends on ordering alone, so nothing there turns on how they are normalised.

2. The frame the determinants are read in

Every empirical number in this paper is a determinant sign or a function of one, and a determinant sign has to be read in a frame that stays conditioned. `I + B` does not. Its condition number grows like `exp(beta * bandwidth)`, and past `beta ~ 4` its eigenvalues stop meaning anything. This section comes first because every table after it runs at a `beta` where that matters.

This is a measured failure (`reads/expBM_naive_frame_manufactures_a_sign.py`, 400 draws a beta). Forming `B` as the plain ordered product and reading its spectrum reports, at **half filling on a bipartite lattice where positivity is provable** and the true negative fraction is therefore zero:

The last two rows are why. The conditioned frame's spectrum stays within three orders across the whole range; the naive product's runs to `5.8e+22`, so its eigenvalues were never near a crossing and its sign is not a reading of one. The naive frame does not merely lose precision; it manufactures the phenomenon the paper is about.

The conditioned frame comes from the UDT factorisation [3]. With `I + U D T = U Db M T` and `prod(Db) > 0`, the sign is carried entirely by `det(U) det(T) det(M)`, and `M` is bounded by construction because `Db >= 1` divides the large block and `Ds <= 1` is the small one.

Three properties hold, and each is checked against a case that must break it (`tests/test_conditioned_frame.py`):

- **two independent routes agree.** The block formulation and the core factorisation reach the sign by different arithmetic and agree on every configuration measured. The two routes must be arithmetically independent for this to mean anything: a parity count compared against `sign(prod(1 + lambda))` from the *same* eigenvalues agrees whatever those eigenvalues are.
- **the answer does not move with the stabilisation block.** Every configuration's sign is compared across three block sizes, not just the totals, so agreement cannot come from cancellation.
- **the naive frame fails.** A check that cannot fail establishes nothing, so the unstabilised product is run alongside and must report the sign problem that is not there.

Everything that follows is read in this frame.

3. The sign is a disagreement between two channels

The weight of an auxiliary-field configuration is

```
w(x) = det(I + B_up(x)) * det(I + B_dn(x))
```

so `sign w(x) < 0` **if and only if** the two channels' determinant signs disagree. That is arithmetic, immediate from the product. What is a result is that the disagreement has structure.

Measured (`reads/expVV_lockstep.py`, 600 configurations per row, 2x4, U = 4, both channels read separately in the conditioned frame of §2; five seeds, errors from the seed spread):

beta	2	4	6	8
negative fraction, naive product	0.0000	0.0000	0.0125	**0.2400**
negative fraction, conditioned frame	0.0000	0.0000	**0.0000**	**0.0000**
largest `|1 + lambda|`, naive	4.4e+06	1.4e+12	6.2e+17	**5.8e+22**
largest `|1 + lambda|`, conditioned	61.1	180.8	113.2	**576.5**

beta	mu	up flips	dn flips	agree	disagree	disagree = neg fraction
4	0.0	0.0020 +- 0.0018	**0.0020**	**1.0000 +- 0.0000**	0.0000	exact
6	0.0	0.0133 +- 0.0046	**0.0133**	**1.0000 +- 0.0000**	0.0000	exact
8	0.0	0.0400 +- 0.0097	**0.0400**	**1.0000 +- 0.0000**	0.0000	exact
10	0.0	0.0753 +- 0.0138	**0.0753**	**1.0000 +- 0.0000**	0.0000	exact
12	0.0	**0.1120 +- 0.0103**	**0.1120**	**1.0000 +- 0.0000**	0.0000	exact
6	0.4	0.0093 +- 0.0025	0.0093	0.9847 +- 0.0027	0.0153	exact
8	0.4	0.0220 +- 0.0040	0.0200	0.9647 +- 0.0079	0.0353	exact
10	0.4	0.0323 +- 0.0035	0.0317	0.9433 +- 0.0070	0.0567	exact
12	0.4	0.0413 +- 0.0077	0.0417	0.9283 +- 0.0049	0.0717	exact

Some columns are estimates and some are not. The two flip rates are proportions of 600 draws and the up column carries its spread across seeds. The agreement is not an estimate: at half filling the two channels change sign on *exactly* the same configurations, so the column reads `1.0000` with zero spread on every seed rather than averaging to it. With the agreement exact, the two flip rates are equal by entailment and the down column repeats the up one; it is printed because on the doped rows, where the agreement is broken, the two come apart. The last column is likewise an identity checked configuration by configuration, not a fitted correspondence – the disagreement fraction **is** the negative-weight fraction, on doped rows as well as half-filled ones.

The half-filled rows establish two things together, and neither alone would do. The individual channels **do** change sign, on around 11% of configurations by `beta = 12`, so the lockstep is not a statement about a quantity that never moves. And they change sign on the same configurations every time, which is what makes the two rates **exactly** equal at half filling – `0.1120` and `0.1120`, and so on every half-filled row to four decimals – while the doped rows are merely close (`0.0220` against `0.0200` at `beta = 8`). The agreement is a half-filling statement and not a generic one: it is `det_dn(x) = det_up(-x)` meeting a field distribution symmetric under `x -> -x`, and doping breaks the second of those.

On four geometries, not one. Every table above is a 2x4 lattice, and the claim is general, so it was remeasured across sizes and shapes (`reads/expAC_geometry.py`, `mu = 0`, `beta = 10`, 300 configurations x **four seeds on every row, controls included**):

The last two rows are controls: a periodic lattice with an odd side has an odd ring and is not bipartite, and there all three claims fail together – the identity breaks, the channels disagree, and negative weights appear. A test that held everywhere would not be sensitive to the property the argument depends on. Every row here is read across four seeds, controls included, because a control read from one seed cannot say whether its failure is the lattice or the draw.

The flip rate is not monotone in `N`*, and the* `E_F` *column is why.* `4x4` flips on `0.4542` of configurations against `0.0050` on the larger `4x6`. For a square lattice with nearest-neighbour hopping the dispersion is `E = -2t(cos kx + cos ky)`, so a state sits at the half-filling Fermi level exactly when `cos kx + cos ky = 0`; `4x4` has **six** such states where every other lattice here has two, and a determinant changes sign when an eigenvalue crosses zero. The count is exact and needs no threshold. What the lockstep claim is about is the *agree* column, and that reads `1.0000` with zero spread across seeds on every bipartite row whatever the flip rate does.

The `4x4` row is therefore the strongest form of the statement available here. An individual channel changes sign on **45.4%** of configurations and the two channels agree on **every one of them**, on all four seeds.

One row is reported and not counted: `4x6` at `beta = 6` has a flip rate of **0.0000 +- 0.0000**, so its perfect agreement is vacuous – nothing crossed. Agreement is only evidence where the channels actually flip, so `tests/test_lockstep.py` requires a non-zero flip rate alongside it.

4. The identity that produces the lockstep

Two facts about the two determinants look incompatible. Their signs agree on every configuration, and their log-magnitudes differ by 8 to 16 orders. A relation `det_up = c det_dn` for constant `c` would hold that log-difference fixed, and it is not fixed – it ranges across configurations, which is the second fact. So the relation cannot be of that form: it must preserve the sign and leave the magnitude free.

It does, and it is exact. The two channels' diagonal factors are inverses of one another, so `det(B_up)/det(B_`

lattice	N	states at E_F	identity residual	flip rate	agree	neg fraction	coupling
2x4	8	2	7.2e-12	0.0817 +- 0.0084	**1.0000 +- 0.0000**	0.0000 +- 0.0000	**-1.0000**
4x4	16	**6**	7.0e-11	**0.4542 +- 0.0241**	**1.0000 +- 0.0000**	0.0000 +- 0.0000	**-1.0000**
2x6	12	2	5.5e-12	0.0508 +- 0.0103	**1.0000 +- 0.0000**	0.0000 +- 0.0000	**-1.0000**
4x6	24	2	9.5e-12	0.0050 +- 0.0019	**1.0000 +- 0.0000**	0.0000 +- 0.0000	**-1.0000**
2x3	6	1	1.4e+01	0.0467 +- 0.0027	0.9042 +- 0.0242	0.0958 +- 0.0242	not resolved
3x4	12	1	7.5e+00	0.0075 +- 0.0074	0.9858 +- 0.0100	0.0142 +- 0.0100	-0.7466

dn) = exp(2 lambda sum x); routing that through det(I + B) = det(B) det(I + B^-1) carries one factor into the ratio of the full determinants:

```
ln|det(I + B_up(x))| - ln|det(I + B_dn(x))| = -dtau * L * tr(K)
+ lambda * sum(x)
```

with lambda = arccosh(exp(dtau U / 2)) the constant of Hirsch's discrete decoupling [2].

Verified (reads/expWW_ratio.py, tests/test_identity.py) at half filling on a bipartite lattice:

The second row is the control: a relation that held for any coefficient would be vacuous. The derived coefficient is exact and the alternative is wrong by thirteen to fifteen orders, the separation narrowing with beta as the derived residual accumulates arithmetic.

Across the interaction, not at one value of it. lambda is a function of both U and dtau, so varying them tests the derivation rather than repeating a measurement: each row predicts a different coefficient and has to be exact at it (reads/expAD_coupling_strength.py, residual max over 200 configurations):

dtau	U = 2	U = 4	U = 8	U = 12
0.0625	2.7e-14	6.8e-13	7.3e-13	4.4e-12
0.125	2.3e-14	6.8e-14	3.7e-11	1.0e-11
0.25	6.0e-14	7.4e-12	1.5e-10	2.8e-08

lambda spans 0.357 to 2.180 across this table, a factor of 6.1. The residual grows gently with it as accumulated arithmetic – the last cell sits at 3e-8, which is nine and a half orders below the wrong coefficient's value on the same row and is float64 rather than a limit of the relation.

And the coefficient is resolved to better than its own variation. Borrowing a neighbouring row's lambda – a difference of as little as 0.51 against 0.74 – fails by 15 to 44 while the predicted one holds at 1e-14:

The relation resolves a 44% difference in lambda by fourteen orders, at every interaction strength measured.

One relation explains both observations. The ratio is an exponential, hence positive, hence the two determinants share a sign – the lockstep, and so the absence of a sign problem. Its exponent is a sum over L * N field components, hence the ratio's magnitude ranges over 8 to 16 orders – so the two determinants' magnitudes look unrelated while their signs are pinned.

The identity is a property of the structure, not of the Ising field. Its derivation uses one property of the decoupling – that the two spins' diagonal factors are inverses, V_up = V_dn^-1. The *continuous* Gaussian spin decoupling has that property too, with its own closed-form constant lambda_s = sqrt(dtau U), so the same derivation predicts the same identity there with a different coefficient. At U = 4, dtau = 0.125 the two constants are 0.73690 and 0.70711 – **4.2% apart** – and neither is fitted (reads/expAK_identity_across_representations.py, 150 draws per cell):

Each representation's own constant is exact to 1e-14 and the other's, four percent away, fails by fourteen orders. A relation that held for any coefficient of roughly the right size could not do this, and neither constant can have been tuned to its data: both are closed forms of their own decoupling, fixed before any determinant is formed.

What the identity carries travels with it. In both representations the two determinants' signs agree on **1.0000** of configurations with no negative weight, and the coupling reads **-1.0000**:

The beta = 4 rows are marked because neither representation flips a determinant sign in 200 draws there, so perfect agreement is agreement about nothing; the rows with content are the deeper ones. §8 shows the read is a property of a (Hamiltonian, decoupling) pair – this is the other half of that statement: within the spin decoupling it does not depend on which field distribution realises it.

5. What the identity requires, controlled

Every row that verified the identity had a bipartite lattice **and** half filling. Those are confounded, and separating them required care: next-nearest hopping tp destroys the bipartite structure, but it also moves the band, so mu = 0 is no longer half filling. Compared without that control the data appears to show that breaking the bipartite structure removes the sign problem – an artifact, because the non-bipartite cells sat at <n> = 0.75.

The filling is therefore fixed by the **non-interacting** density, a closed-form function of the single-particle spectrum that carries no sampling noise and is computed without touching the sign.

Measured (reads/expZZ_matched_filling.py, U = 4, beta = 8, every row at n = 1.00000):

The tuner recovers mu = -0.0000 for the bipartite lattice without being told that half filling lies there, which is the control validating itself. At identical filling, the bipartite lattice has no sign problem and the identity is exact; every non-bipartite one has a sign problem and the identity is broken.

beta	2	4	6	8
residual, coefficient `lambda`	**1.51e-14**	**4.35e-14**	**1.08e-12**	**2.09e-12**
residual, coefficient `2 lambda`	2.51e+01	3.83e+01	5.31e+01	4.86e+01

U	own lambda	borrowed	residual, own	residual, borrowed
2	0.51048	0.73690	2.3e-14	**15.4**
4	0.73690	1.08504	9.6e-13	**18.8**
8	1.08504	1.38202	1.3e-12	**17.8**
12	1.38202	0.51048	5.1e-11	**43.6**

field	beta	its own lambda	the *other* representation's lambda	twice its own
Ising	2	**1.1e-14**	9.5e-01	2.4e+01
Ising	4	**3.1e-14**	1.4e+00	3.4e+01
Ising	6	**2.8e-12**	1.5e+00	3.8e+01
Gaussian	2	**3.7e-14**	8.8e-01	2.1e+01
Gaussian	4	**3.6e-14**	1.2e+00	2.9e+01
Gaussian	6	**4.3e-14**	1.5e+00	3.5e+01

For a real one-body matrix the two conditions are one criterion, checkable without sampling. The criterion below is a statement about the spin decoupling, as everything in §§3-7 is: it is derived from that decoupling's diagonal factor and says nothing about the charge channel, whose positivity comes from a different mechanism (§7). The derivation reduces to §4's form only when `det(I + B_up^-1) = det(I + B_dn)`, which needs `expmK^-1` similar to `expmK` – and the similarity has to commute with the interaction's diagonal factor. That restricts it to a **signed diagonal**: `S K S = -K` with `S = diag( +-1)`. Elementwise `s_i s_j K_ij = -K_ij`, so the criterion is a 2-colouring of K's support graph together with a zero diagonal. It costs `O(N^2)`, and uses no eigenvalues, no determinants and no field (`reads/expBE_criterion_tables.py`, 2x4, `U = 4`, `beta = 8`):

Twelve one-body matrices, agreeing with the measured identity on every one. The criterion unifies §5's two conditions and covers a case neither names: a **staggered potential** leaves the lattice bipartite and the filling at exactly one per site, and breaks the identity anyway, because a diagonal term cannot be negated by signs.

The bond-disordered rows are the ones that make it predictive rather than descriptive. No symmetry was designed into them – every bond strength is drawn at random, up to a spread of 1.5 – and the criterion says the identity holds, which it does to `1e-13`.

It is a statement about the hopping graph, not about lattices. The same criterion was applied to topologies the rest of the paper never uses, with the prediction made before the measurement:

topology	criterion	identity residual
chain, 8 sites, periodic	yes	3.7e-13
chain, 7 sites, periodic	no	6.3e+00
chain, 8 sites, open	yes	2.0e-13
star graph (a tree), 8 sites	**yes**	**6.8e-13**
triangular ladder, 8 sites	no	1.0e+01
even ring, 12 sites	yes	6.5e-13
odd ring, 9 sites	no	5.9e+00

A tree satisfies it however it is drawn, having no cycles at all; any graph carrying an odd cycle never does. Seven more topologies, agreeing with the measured identity on every one.

With complex hoppings there are two routes, and the identity holds when either is open. The signed-diagonal criterion is sufficient but not necessary once `K` is complex: a 5-site ring threaded by a flux of `pi/2` has no signed-diagonal conjugation and satisfies the identity anyway, at `1.4e-14`. The conjugation may be a diagonal **unitary** rather than a diagonal sign, and the derivation is satisfied by either a similarity or an **anti**-similarity:

route	condition	consequence
A	`S K S^-1 = -K`	every cycle **even**, at any flux
B	`S K S^-1 = -conj(K)`	`(-1)^l e^{-2i Phi} = 1` on every cycle: **odd** cycles need flux `pi/2 mod pi`

An odd cycle carrying half-odd-integer flux restores what the odd cycle destroys. Measured on rings of 5, 6, 7 and 8 sites at six fluxes each, plus a 5x2 lattice at five fluxes (`reads/expBA_oracle_superset.py`) – **29 of 29 agree**, including every case where exactly one route is open.

A phase condition on the cycles of a graph built from the Hamiltonian is a known shape. Hen [18] gives a criterion of exactly that form for quantum Monte Carlo generally: the simulation is sign-problem-free precisely when

field	beta	flip rate	signs agree	negative weight	strength	agreement is
Ising	4	0.0000	1.0000	0.0000	-1.0000	*vacuous, no flips*
Ising	8	0.0350	**1.0000**	0.0000	**-1.0000**	real
Ising	12	0.1450	**1.0000**	0.0000	**-1.0000**	real
Gaussian	4	0.0000	1.0000	0.0000	-1.0000	*vacuous, no flips*
Gaussian	8	0.0200	**1.0000**	0.0000	**-1.0000**	real
Gaussian	12	0.0900	**1.0000**	0.0000	**-1.0000**	real

tp	tuned mu	bipartite	identity residual	signs lock	neg fraction
0.00	**0.0000**	yes	**4.4e-13**	**1.0000**	**0.0000**
0.15	0.6000	no	4.25e+01	0.9500	0.0500
0.30	1.1992	no	8.32e+01	0.8150	0.1850
0.70	1.5548	no	1.05e+02	0.9200	0.0800

K	signed-diagonal S exists	identity residual	agrees
bipartite, `mu = 0`	yes	2.4e-12	yes
bipartite, staggered `h = 0.2`	**no**	5.7e+00	yes
bipartite, staggered `h = 1.2`	**no**	2.0e+01	yes
`tp = 0.3`	no	1.4e+01	yes
`mu = 0.4`	no	3.2e+01	yes
bipartite, bond-disordered x3	**yes**	5.2e-14 to 1.2e-12	yes
4x4 and 2x6 bond-disordered	**yes**	2.1e-12, 6.4e-13	yes
odd rings of 5 and 7 sites	no	7.3, 6.3	yes

the total phases along the chordless cycles of the weighted graph whose adjacency matrix is the Hamiltonian vanish. Route B is the same shape one level down – the graph here is the support graph of the one-body matrix `K`, not of the Hamiltonian, and what the condition decides is the pairing identity of §4 rather than positivity directly, which is why §5 separates the two and finds cases where the identity holds and positivity does not.

For real `K` the two conditions coincide, which is why the signed-diagonal colouring is the whole story there: checked on 11 real matrices, the general criterion and the colouring return the same answer on every one.

The tables above overlap – a bond-disordered `4x4` appears in one and a `4x4` flux row in another – so they are not added up here. `reads/expBA_oracle_superset.py` enumerates the population once, independently of how the sections group it: **69 one-body matrices, no mismatch**, of which the flux-route subset is **29 of 29**.

The two routes are not equivalent, and only one carries positivity. Route A leaves the weights real: on `ring 6` and `ring 8` at flux `pi/4` the identity holds at `2e-14` and the weight's mean phase is **exactly 1** – no sign problem. Route B restores the identity on an odd cycle and leaves a phase behind. Measured with `1 - |<w/|w|>|`, which is the ordinary negative-weight measure when the weight is real and the mean-phase deficit when it is not (`reads/expBE_criterion_tables.py`, `beta = 8`, 200 draws):

How many digits of a sign deficit are real. The deficit is `1 - |<w/|w|>|`, a sampled mean over draws, so it carries that mean's error and each table quotes its own run. Measured across six seeds (`reads/expBG_single_channel_has_no_indicator.py`): at 200 draws the `ring 5` deficit spans `0.14457` to `0.17446` with a standard deviation of `0.01059`, and the `tri ladder 8` deficit spans `0.44611` to `0.54008` at `0.03284`; at 800 draws both spreads roughly halve, to `0.00707` and `0.01918`. So a deficit quoted here is reproducible in its second decimal, and two sections quoting the same lattice from different runs differ by that much and not by more.

The last two rows show the two coming apart. Threading the frustrated ladder does what the criterion says – the identity goes from `1.2e+01` to `4.3e-14` – and the sign deficit rises from `0.19000` to **`0.50071`**. The flux converts a mild sign problem into a substantial phase problem.

So the criterion decides **the identity**, and the identity delivers **positivity only where the weights are real**, which is route A. On an odd cycle no flux-free choice exists and route B is the only one available, so an odd cycle cannot be made positive this way at all – it can only be made to satisfy the identity.

That separation is not local to this model. A pairing identity holding while positivity fails is the same distinction reflection positivity turns on, where "the measure is invariant under the reflection" is the easy half and the content is whether the weight can be written as a pairing at all. They coincide on the well-behaved cases and come apart on route B over an odd cycle: the identity exact at `4.3e-14`, the sign deficit `0.50071` against `0.19000` unfluxed.

On a two-dimensional frustrated lattice the answer depends on the size, and the criterion tracks it. A periodic triangular lattice has triangles, rhombi

K	route	identity residual	sign deficit
2x4 real, half filled	A and B	2.4e-12	**0.00000**
ring 6, flux `pi/4`	**A only**	2.8e-14	**0.00000**
ring 5, flux `pi/2`	**B only**	2.0e-14	0.14968
triangular ladder, flux 0	neither	1.2e+01	0.19000
triangular ladder, flux `pi/2`	**B only**	4.3e-14	**0.50071**

and wrap-around cycles, and route B has to hold on all of them at once. On `4x4` the wraps are even and carry no flux, so `theta = pi/2` opens route B and the identity holds to `4.7e-13`; on `3x3` the wraps are odd and flux-free, no `theta` opens either route, and the best residual over five fluxes is `3.2` – the identity stays broken at every one. Ten rows across the two lattices and five fluxes, all predicted correctly (`reads/expBO_triangular_pair.py`).

What the pair shows. It is a statement about the **identity**. The unfluxed `4x4` carries a broken identity – residual `9.26` at `beta = 8`, rising to `14.59` at `beta = 16` – and still shows **no negative weight at all** in 200 draws at either, so nothing was restored there that was measurably absent. The `3x3` lattice does have a real sign problem, `0.3500` unfluxed and `0.1351` at `pi/2`, and the criterion correctly says no flux removes it. This pair extends the criterion to two dimensions and shows the wrap cycles deciding the outcome; like the ladder, it is a statement about the identity and not about positivity, since `theta = pi/2` opens route B.

The obvious criterion is the spectral one, and it is wrong. For Hermitian `K`, a spectrum symmetric about zero is exactly similarity to `-K`. The staggered rows have spectra symmetric to **1e-15** and break the identity by up to `21`. The similarity exists; it is not a signed diagonal, so it does not commute with `diag(exp(sigma lam x))` and the derivation cannot use it.

The criterion is sharp: the identity is fragile where the sign problem is robust. A staggered term of any amplitude `eps > 0` makes `S K S = -K` unsatisfiable, and the identity follows exactly (`reads/expBE_criterion_tables.py`, 100 draws a row):

The residual is **linear in the perturbation**, with no tolerance and no onset: the identity does not survive a small violation, it degrades in proportion to it from the first. The ratio holds to `0.07%` across three decades of `eps`, from `1e-06` to `1e-03`, and is `7.5%` high by `1e-02`. A chemical potential behaves the same way, at a ratio of `92.4` across three decades.

The negative fraction does not follow suit. It is exactly `0.0000` until `eps` reaches about `0.1`, so a model can violate the criterion, carry a measurably broken identity, and still produce no negative weight at all. **The identity is structurally fragile and the sign problem numerically robust**, and those are statements about two different quantities – the residual measures the perturbation, the negative fraction measures its consequence, and only the first is linear.

So the criterion decides the **identity**, and nothing further on its own: it carries positivity only along route A, where the weights stay real, and it does not predict the negative *fraction* on either route. That is the mechanism behind the next point.

The residual is not a severity measure (`reads/expAB_coupling_controlled.py`). It grows monotonically with `tp` while the negative fraction does not: on that experiment's own draws it runs `0.0375`, `0.1825`, `0.0600` across `tp = 0.15, 0.30, 0.70`, rising and then falling. The matched-filling table above (`reads/expZZ_matched_filling.py`) measures the same axis on its own draws and finds the same shape at `0.0500`, `0.1850`, `0.0800`; the two are separate samples of one quantity and agree on the ordering, which is the claim. The identity's failure permits the signs to differ; it does not say by how much they will.

The scope of this axis. The rows above are an `if and only if` on it: at matched filling, bipartite means the identity holds and there is no sign problem, and non-bipartite means both fail. Positivity has other sources. Wei et al. [7] give a unified account of the known sign-free models and identify sign-free models with *repulsive* interactions and *without* particle-hole symmetry, and Wu and Zhang [6] derive positivity from a conjugate-pairing property tied to time-reversal rather than to a lattice bipartition. §7 measures one of those other places directly – the attractive charge channel, sign-free at every filling – and finds the read saturated there too, at the opposite sign. So the symmetric point is one source of positivity that this rig can read, and the scope of §5 is the axis it was measured on.

The criterion above is the standard one. That the repulsive Hubbard model is sign-free at half filling on a bipartite lattice has been known since the method's early years, and the usual statement of the mechanism – the two spin determinants related by particle-hole symmetry – is what §4's identity writes as an equality and §5's `S K S^-1 = -K` writes as a condition on `K`. [6] and [7] give the modern account of that landscape and are cited above for it. For real `K` the two-colouring is bipartiteness and the zero diagonal is the absence of a staggered potential, so the agreement on 69 of 69 one-body matrices is a check that this rig computes the known criterion correctly. What the rig is for begins at §6: the same verdict taken from the weights a simulation already holds, with no `K` in hand – a question about what is recoverable from data.

eps	criterion	identity residual	residual / eps	negative fraction
0	yes	3.2e-13	–	**0.0000**
1e-06	no	8.542e-05	**85.425**	**0.0000**
1e-04	no	8.543e-03	**85.425**	**0.0000**
1e-03	no	8.548e-02	**85.482**	**0.0000**
1e-02	no	9.185e-01	91.848	**0.0000**
1e-01	no	3.217e+00	32.165	0.0800

6. The criterion performed on output, with no Hamiltonian

Everything above decides the identity from `K`: build the support graph, look for a diagonal unitary with `S K S^-1 = -K`, two-colour it, check the flux on every odd cycle. That is a decision about a matrix. A running simulation does not have a clean matrix to hand – it has configurations and determinants – and the terms most likely to have broken the identity, a doping term or an added staggered potential, are exactly the ones a criterion is rarely rederived for.

Read as a statement about the data rather than about `K`, §4's identity says something a read can act on. Across configurations,

```
ln|det_up|(x) - ln|det_dn|(x)   is exactly affine in   sum(x)
```

with slope `lambda` and offset `-dtau L tr(K)`. That is one channel against another on a shared index, which is what `coupling` reads, and the read is invariant to both the offset and the scale – so neither constant has to be known. An exact affine relation must saturate a normalised alignment at 1. A broken identity cannot (`reads/expAW_criterion_from_output.py`, 2x4 and rings, `U = 4`, `beta = 6`, 400 configurations):

	`1 - |strength|`
identity holds (8 lattices)	`0` to `2.2e-16` – machine epsilon
identity fails (7 lattices)	`1.0e-3` to `2.9e-2`

The read agrees with the measured identity residual on 15 of 15 lattices, and with §5's algebraic criterion on 15 of 15, using no Hamiltonian. Neither side is given the other's input: `coupling` sees two columns and never sees `K`; the criterion sees `K` and never sees a configuration. The agreement is counted against a cut on `1 - |strength|`, and the two populations are thirteen orders apart – `2.2e-16` against `1.0e-3` – so every cut in that gap returns the same 15 of 15. No cut is chosen here because none of them differ.

The departure is systematic rather than sampling noise. Across a factor of four in sample size and three seeds, the saturated lattices stay at machine zero and the broken ones stay at `9.8e-4` to `1.4e-2` without shrinking. A departure that fell as `n` grew would be the read reporting its own variance.

What it costs and what it is for. Two columns a simulation already has, and one read – `O(n)` after determinants that were computed anyway, against a criterion that otherwise needs the Hamiltonian's structure. It turns "does this model still satisfy §4's identity" from a question asked once on paper into a check a run can perform on itself.

6.1 How it is used

Log two scalars per configuration. Both are already computed by any determinant QMC code:

- `ln|det_up| - ln|det_dn|`, which comes out of the stable propagation – `slogdet` is formed every sweep to build the weight;
- `sum(x)`, a reduction over the auxiliary field.

Then one call, `coupling(A, B)`, and read whether it saturates. Three places to do it:

1. **As a pilot, before committing to a production run.** A few hundred configurations at small `beta` decide whether §4's identity holds for the model *as assembled*. That is the identity and not positivity: on §5's route B the identity holds and the weights still carry a phase, and §7.1 measures four such lattices where the read is exactly saturated and the sign deficit runs from `0.08482` to `0.36442`. A saturated pilot rules out a broken pairing, not a sign problem.
2. **As a regression check.** A sign-free setup acquires a term – `t'`, a staggered field, a boundary twist, a chemical potential. The two-colouring is rarely rederived when a term is added; this catches it from the output.
3. **As a build check.** The read sees what the code *did*, not what the model was meant to be. Where an assembly routine and its documentation disagree, the read follows the code.

6.2 The build check, measured

The third use is the one the other two cannot cover, because it is the case where the criterion of §5 is *unavailable by construction*: if the code may not be running the model its author believes, then the `K` and the `lambda` one would feed to a direct test of §4's identity are the suspect quantities themselves. The read needs neither – it is invariant to the offset and the scale – so it stays available (`reads/expAZ_build_check.py`).

Intended: an **open** 7-site chain, bipartite, and §5's criterion computed on it reports sign-free. As built: a periodic wrap, making a 7-site **ring** – an odd cycle, outside the class. The oracle is silent for every row below, because it reads the model rather than the code.

beta	negative fraction (4 seeds)	1 - |strength|	seed spread	correct-build control
1	**0.00000**	1.904e-07	2.1e-09	1.1e-16
2	**0.00000**	6.319e-05	5.8e-06	-2.2e-16
3	**0.00000**	3.336e-04	6.1e-05	1.1e-16
4	0.00125	7.381e-04	1.0e-04	0.0e+00

At beta = 1 to 3 not one negative weight appears across four seeds, so every standard health check passes, while the read stands eleven orders above the correctly-built control at beta = 2 and between 5.5 and 91 times its own seed spread. From beta = 4 the sign fires too and the read is no longer needed.

The scope has two provable limits: one blind spot, and one degenerate input the read reports. A wrong decoupling constant – the continuous Gaussian sqrt(dtau U) used in the discrete field – is **not** detected: 1 - |strength| is -2.2e-16, exactly saturated, because rescaling an affine relation leaves it affine. Scale invariance is why the read needs no lambda and why it cannot see a wrong one; the two are the same fact, and the direct residual catches that case at 1.25, so the two checks are complementary. Applying the field with one sign to both channels leaves the two determinants identical, so the logged difference is identically zero and the read returns **unresolved** rather than a departure – a degenerate input, visible before any read is taken.

6.3 Compared with watching the average sign

Every DQMC code already reports <sgn>. The read fires earlier. On 2x4 with t' = 0.3, at beta = 6 over 400 configurations:

quantity	value
negative-weight fraction	**0.0000** – not one negative weight
§4 identity residual	**10.53** – broken by an order of magnitude
1 - |strength|	**2.3e-3** – flagged

The average sign reports a clean run on a model that has already left the sign-free class. The same holds on ring 5 (deficit 0.0167, identity broken by 6.26) and on both staggered lattices.

<sgn> is a *sampled* quantity: it sits at 1.0000 until the sign problem grows large enough to appear in a finite sample, and the cost of it appearing is O(1/<sgn>^2) with <sgn> = exp(-beta N df). So the expense of discovering the problem that way rises exponentially in beta and in system size, while this read's cost is flat in both – O(n) over two columns, at whatever beta is cheapest to sample. What it buys is the difference between learning at small beta, where a pilot is minutes, and learning at production beta from error bars that will not shrink.

7. Reading the weights: which problem is it, how far from the symmetric point, and at what cost

An observer standing at the output holds a cloud of configuration weights and nothing else – no K, no knowledge of the decoupling, and no constant to supply. This section is what can be decided from there, and it has three parts, in the order a practitioner meets them.

First, which problem the run has. concentration on the weight cloud's (Re, Im) frame returns a *directional* statistic and an *axial* one, and only the pair classifies: focus = 1 with resultant = 1 is sign-free; focus = 1 with resultant < 1 is a real sign problem, because rank one in the plane means the phase takes two values pi apart and a phase confined to Z2 is what a sign *is*; focus < 1 is a phase problem no rotation reaches. That triage is §7.1, and it decides what the rest of the section measures.

Second, how far the run sits from the protecting symmetry, which is the coupling read below – exactly calibrated where the lockstep is perfect, and moving while the average sign is still identically 1.

Third, what any of it can cost, which §9 bounds.

The coupling read that follows is a statement about a *real* determinantal weight, so the classification comes first.

7.1 The triage: which problem is it?

The triage recovers §5's positivity verdict from the weights alone (reads/expAV_monomial_conjugation.py). §5 establishes from K that route B buys the identity *without* positivity. That same verdict is readable from the output, where an observer actually stands – holding a finite sample of weights and no knowledge of the mechanism. On four route-B lattices the identity holds to machine precision, so every comparison between the channels is saturated and blind (beta = 6, 120 draws – a shallower point and a smaller sample than §5's table, so the deficits are not the same numbers):

Every row satisfies the identity and still carries a phase. focus below 1 says the cloud is not rank one, so no global rotation makes those weights real – which is §5's statement that route B carries the identity without positivity, obtained with no K and no mechanism. Nothing here is thresholded: focus is compared to 1, the definitional value of a rank-one cloud.

This is the case that separates the two halves of §7. The channel comparison of §7.2 is exactly saturated on all four rows and reports nothing; the one-sided triage reads the answer off the cloud's own geometry.

lattice	§4 identity residual	focus	|Im| after de-rotation	sign deficit
ring 5, flux pi/2	2.04e-14	0.75254	1.000	0.10998
ring 7, flux pi/2	2.22e-14	0.83339	1.000	0.08482
tri ladder 6, flux pi/2	2.13e-14	0.54294	1.000	0.31001
tri ladder 8, flux pi/2	3.91e-14	0.57340	1.000	0.36442

The same quantity read one-sidedly is available. On the weight's unit-modulus frame, concentration. resultant – the length of the mean row, the von Mises-Fisher sufficient statistic – **is** |<w/|w|>| by construction rather than by measurement, so its agreement is arithmetic and not a finding. What it supplies is the value, on rows the comparison could not reach at all:

K	|<phase>|	resultant
2x4 clean	1.00000	**1.00000**
ring 6, flux pi/4	1.00000	**1.00000**
staggered h = 0.6	0.93500	**0.93500**
ring 5, flux pi/2	0.84133	**0.84133**
tri ladder, flux pi/2	0.53586	**0.53586**

And the companion read separates the two failure modes (reads/expAU_axial_versus_directional.py). concentration reports a directional statistic and an axial one, and they differ exactly where it matters: an antipodal cloud reads resultant ~ 0 and focus ~ 1. A real sign problem *is* antipodal – the weights sit at +-1, on one line through the origin – while a phase problem is spread around the circle. Measured, focus is **1.0000** on every real-weight row of that experiment and 0.5734 to 0.7525 on its three genuinely complex ones; across the complex rows of §7.1's table (reads/expAV_monomial_conjugation. py) as well it runs 0.543 to 0.833.

focus = 1 is rank one in the (Re, Im) plane, which is the statement that **a single global rotation makes every weight real**. The rotation is read off the cloud's own leading direction, no angle chosen, and applying it returns a real sign problem carrying the same |<sgn>|: rows rotated by 0.7 and 1.9 present a maximum imaginary part of 0.64 and 0.95, de-rotate to 9e-16 and 4e-16, and recover 0.93500 and 0.92500 exactly. Where focus < 1 no rotation helps and the residual stays at 1.0.

A global phase cancels. A phase common to every configuration multiplies numerator and denominator of <O> = sum(O w)/sum(w) alike, so it cancels exactly and costs nothing. Measured: rotating the weights by 0.3 to 1.3 leaves the mean phase |<u>| at **0.93500** throughout, unmoved (reads/expBN_a_global_phase_and_ the_wrong_estimator.py; u = w/|w|, the frame resultant is read on).

It is only visible to an estimator that takes the real part. mean(Re u)/mean(|u|) – which is correct for real weights and is the natural thing to carry over – reads cos(theta) too small, falling from 0.93500 to 0.25011 across those same rotations, and would overstate a cost that goes as 1/<sgn>^2 by **13.98** times. focus therefore identifies when the real-part estimator has stopped being the right one.

What that leaves. focus and resultant answer different questions, and only the pair classifies. focus = 1 says the cloud is rank one in the (Re, Im) plane, so a single global rotation makes every weight real – it does **not** say the run is sign-free. What survives that rotation is a real problem whose severity is resultant: 1.00000 on the clean lattice and on the inert twist, 0.93500 and 0.92500 on the staggered and doped rows, which are sign problems and are exactly the ones §5's criterion speaks to. focus < 1 says no rotation reaches it and the phase varies configuration to configuration: it does not cancel in the ratio, and it is the problem itself. So the three regimes are (focus = 1, resultant = 1) sign-free, (focus = 1, resultant < 1) a real sign problem, and focus < 1 a genuine phase problem. The read separates a representation artefact from the physics, which is a diagnosis and not a saving.

The sign problem is the rank-one case of a phase problem. Rank one in the plane is the statement that the phase takes exactly two values pi apart – a Z2 subgroup of U(1) – and a phase confined to Z2 is what a sign *is*. So focus reads the rank, the rank decides which of the two a run has, and both are read from the weights alone with no constant supplied and no Hamiltonian consulted.

So a caller with complex weights can ask which problem they have before deciding what to do about it: focus = 1 says a phase that a rotation removes, leaving the sign problem §5's criterion speaks to; focus < 1 says a phase that no rotation removes, which is what route B produces.

So the phase is readable, and the blindness above is a statement about *comparisons* rather than about what can be measured. The two halves fit together: a common mode is invisible to a read of the relation and visible to a read of one side. And what the one-sided read returns is |<sgn>| itself, so it carries the ceiling of the paragraph below – seeing the phase exactly and paying the same O(1/<sgn>^2) to resolve it.

A second collision, with the channels exactly related rather than identical. §5's route B restores the identity on an odd cycle by threading it with flux, and the two channels then satisfy G_dn = 1 - G_up to machine precision – while the weight carries a phase. Four systems whose channel relation is exact and whose sign behaviour is not (reads/expBD_route_calibration.py, beta = 8, 250 draws; the deficit is 1 - |<w/|w|>|):

The relation is exact on every row, both reads return the same saturated value on every row, and the sign deficit runs from 0 to 0.478. **No read of the relation**

K	relation residual	sign deficit	coupling
2x4 real, half filled	1.1e-11	**0.00000**	-1.0000
ring 5, flux `pi/2`	**7.7e-15**	**0.15581**	-1.0000
ring 7, flux `pi/2`	**5.0e-15**	**0.09241**	-1.0000
triangular ladder, flux `pi/2`	**1.2e-14**	**0.46110**	-1.0000

between the two channels can see a phase problem, because the relation does not carry it: `G_dn = 1 - G_up` holds identically whether the weight is positive or spread across the circle.

That is a second impossibility of the same kind as the one below and it is not the same statement. Below, the two channels are identical *as data*, so nothing remains to compare. Here they are distinct and exactly related, and the comparison is well posed and still blind.

The other direction is an impossibility, not a limitation. In the control the two channels are bit-identical, and three systems make that concrete – `numpy.array_equal(A, B)` is `True` in every row:

The read returns the same saturated value on all three while the negative fraction spans 33 percentage points, so its output does not determine the sign behaviour. And when `B = A` the pair carries no between-channel content whatsoever: every statistic of the pair collapses to a statistic of one channel. That is a limit on **any** comparison between channels, not on this read. The control's sign problem sits in the phase of `det(I + B_up)`, a global property of the determinant, while the frame the read is given is the Green's function's diagonal. Whether a read of a single channel could recover that phase is not measured here.

Which twists are live. The total phase round the x-cycle is `Lx * phi`, so a twist that is a multiple of `pi` is gauge-equivalent to a real boundary condition and the determinants stay real; on `Lx = 4` that makes `pi/4` and `pi/2` inert. A periodic lattice of width 2 is inert at every `phi`, because its x-direction has one distinct bond per row reached from both ends and the two Peierls phases add to `-2t cos(phi)`. Every row therefore reports `max |Im G|`, and the width is asserted, so a row cannot be read unless its flux is doing something.

Three quantities have now been checked for whether they measure severity – the identity residual (§5), the coupling strength, and the coupling tightness – and none does. The severity is the negative fraction, and the negative fraction is a count of parity disagreements. §8 gives the reason none of the three reaches it: each has already summed that count away.

7.2 The coupling read, and its exact calibration

The lockstep of §3 is produced by the identity of §4, and the identity holds exactly at the symmetric point of §5. That suggests a continuous measure of how far a system sits from that point, and there is one: the alignment between the two channels, read as a coupling.

The calibration is exact. At half filling on a bipartite lattice, particle-hole symmetry gives `G_up[i,i](x) + G_dn[i,i](x) = 1` configuration by configuration – verified over 2000 configurations a row at a maximum residual of **4.9e-15** at `beta = 2`, rising to `1.2e-12` at `beta = 8` (`reads/expTT_order_parameter.py`) – so the two centred channels are exact negatives and their coupling must read `-1`. It reads **-1.0000**, at every beta (`reads/expQQ_coupling_vs_sign.py`), with the instrument's own exact re-pairing null placing the permuted control at `|z| <= 1.7` (`reads/expAL_capability_across_representations.py`). On the controlled axis of §5 it reads `-1.0000` with `tightness = 1.000` (`reads/expAB_coupling_controlled.py`): the entire coupling in a single mode.

What the deficit measures, exactly. The read is a normalised alignment, so its deficit from saturation is an angle:

```
deficit = 1 - cos(angle between the two centred channel frames)
```

Write the violation of the particle-hole relation as `E = B~ + A~`, and split it into the part along `A~` and the part perpendicular to it. Two consequences follow, and both are measured:

Both are measured on this section's own chains, at the same model, betas and seeds as the capability table below (`reads/expBH_perpendicular_violation.py`):

- **The parallel part is invisible.** Rescaling one channel takes `|E|/|A~|` from `0` to `1` to `4` – `B = 1 - A`, then `1 - 2A`, then `3 - 5A` – and leaves the deficit at **exactly** `0.00000000` at every one. The read has no notion of the two channels' relative size, only of their alignment.
- **The perpendicular part is what registers**, and to leading order `deficit = |E_perp|^2 / (2 |A~| |B~|)`:

The quadratic form is the leading term of `1 - cos`, so it is close where the angle is small and low once it is not: the miss grows monotonically from `1.68%` to `13.92%` as the deficit grows by `5.6x`. Both columns come from the same run, so the relative miss is exact within it and carries the digits shown; the deficit's own reproducibility across seeds is the capability table's (`reads/expAN_capability_table.py`) – `0.6%` at the top and `10.4%` at `beta = 6`, so the last row's deficit is `0.20` as a measurement, not `0.20127`. Past the shallow rows the exact cosine is what holds, and the exact cosine is what the read computes. The deficit column is the capability table's own, which is what makes the comparison a statement about these chains rather than about a synthetic frame.

system	B == A	negative fraction	strength	z	resolved
attractive, no flux	**True**	**0.0000**	1.0000	54.5	yes
control, phi = pi/16	**True**	**0.0300**	1.0000	57.1	yes
control, phi = 3pi/16	**True**	**0.3333**	1.0000	60.3	yes

beta	deficit	|E_perp|^2 / (2|A~||B~|)	relative miss
1.0	0.03574	0.03514	**1.68%**
1.5	0.04893	0.04769	2.55%
2.0	0.06141	0.05956	3.01%
3.0	0.10788	0.10135	6.06%
4.0	0.12512	0.11618	7.14%
6.0	0.20127	0.17326	**13.92%**

So the order parameter is the **angle between the two centred channel frames**, and what it tracks as filling or temperature moves is the growth of the component of the particle-hole violation that cannot be absorbed into a rescaling. That is why it is an onset detector: the angle moves as soon as the violation appears, which is before <sgn> has left 1.

The calibration comes from one of §5's two routes, not from the identity. §5 shows the identity holds when either S K S^-1 = -K (route A) or S K S^-1 = -conj(K) (route B) is available. Only route B gives the exact particle-hole relation, and with it the exact -1 (reads/expBD_route_calibration.py, 250 configurations per row; the criterion itself is reads/expAO_spectral_criterion.py, and the separation is gated in tests/test_spectral_criterion.py):

Route B is the anti-similarity, and it is what forces G_dn = 1 - G_up configuration by configuration – the identity this section's calibration rests on. Where only route A is open the lattice is still sign-free and the identity of §4 still holds exactly, but the two channels are no longer exact negatives about 1/2, and the read comes back at -0.945 to -0.993: resolved, close, and not calibrated.

Each route supplies its own exact relation between the channels. Route B gives G_dn = 1 - G_up; route A gives the same thing composed with complex conjugation, G_dn = 1 - conj(G_up). Both hold configuration by configuration, to 5e-15 and 3e-15 respectively. Under route A the real parts are therefore exact negatives about 1/2 and the imaginary parts are exact *positives*, so the two halves of the frame read -1 and +1 and a single signed alignment averages them.

That average is not approximate. Writing r for the imaginary part's share of the centred variance, the read is

```
strength = -1 + 2r
```

and it holds to **4.4e-16** across six flux-threaded rings (reads/expBL_route_A_is_derived.py; every row is route A only, and sign-free):

K	r	-1 + 2r	measured
ring 6, flux pi/8	0.00091	-0.99817	**-0.99817**
ring 6, flux pi/4	0.00374	-0.99251	**-0.99251**
ring 6, flux pi/3	0.00683	-0.98634	**-0.98634**
ring 8, flux pi/4	0.02736	-0.94529	**-0.94529**
ring 8, flux pi/3	0.02215	-0.95570	**-0.95570**
ring 10, flux pi/4	0.00216	-0.99568	**-0.99568**

The departure from -1 under route A is the imaginary weight of the frame, exactly; the sign problem is absent throughout.

-1.0000 therefore comes from the particle-hole structure route B supplies, which the half-filled bipartite case happens to have – not from positivity, and not from the §4 identity. The three coincide in every measurement in §§3-5 and come apart under flux.

It is readable where the average sign is not. Both columns come from the same importance-sampled chains at each beta (reads/expAN_capability_table.py; StableChains is gated against brute-force enumeration of every auxiliary field at 5e-15, which reads/expQQ_coupling_vs_sign.py states at the head of its own output; N = 8, U = 4, dtau = 0.125, t2 = 0.7, mu = 1.0, four seeds, errors from the seed spread):

The first two rows are the capability. <sgn> is an average of a binary: where the channels are locked it is identically 1 with zero variance and no derivative. The coupling has already moved: 0.03574 +- 0.00022 across four seeds, which is 162 times its own seed spread away from saturation, on a row where <sgn> has no derivative at all.

What |z| is, and what it is not. |z| is the instrument's own resolution against its exact re-pairing null, and that null is centred at strength = 0. For a single pair of columns it is exactly |strength| sqrt(n - 1) – pinned at nine settings in tests/test_the_reads_are_the_statistics_named.py – so it answers whether the two channels are coupled at all, which is not the claim of this section. It is reported because resolved gates on it: an unresolved read returns strength = 0 rather than a number. It scales with the sampling budget rather than with the physics – 50 at R = 24, n_meas = 15 against 137 at R = 64, n_meas = 40 on the same rows – and it falls down the

K	route A	route B	max|G_up + G_dn - 1|	strength
2x4 real	yes	yes	1.1e-11	**-1.0000**
ring 8, real	yes	yes	4.6e-13	**-1.0000**
ring 6, flux pi	yes	yes	1.5e-10	**-1.0000**
ring 5, flux pi/2	no	**yes**	**7.7e-15**	**-1.0000**
ring 7, flux pi/2	no	**yes**	**5.0e-15**	**-1.0000**
triangular 4x4, flux pi/2	no	**yes**	**4.5e-12**	**-1.0000**
ring 6, flux pi/4	**yes**	no	**1.4e-01**	-0.9925
ring 6, flux pi/3	**yes**	no	**1.9e-01**	-0.9863
ring 8, flux pi/4	**yes**	no	**6.4e-01**	**-0.9453**

beta	<sgn>	coupling deficit	reproducibility	|z|
1.0	**1.00000 +- 0.00000**	0.03574 +- 0.00022	0.6%	136.9
1.5	**1.00000 +- 0.00000**	0.04893 +- 0.00035	0.7%	134.0
2.0	0.99863 +- 0.00037	0.06141 +- 0.00039	0.6%	131.3
3.0	0.94961 +- 0.00535	0.1079 +- 0.0033	3.1%	123.2
4.0	0.81836 +- 0.01234	0.1251 +- 0.0019	1.5%	118.7
6.0	0.48633 +- 0.00733	0.201 +- 0.021	10.4%	106.5

table because |strength| falls, which is the deficit rising. The claim that the deficit has moved is carried by the seed spread in the table, not by |z|.

The re-pairing null assumes exchangeability, and these are Markov chains. §9.2c measures 2 tau for the influence function at 2.60x at beta = 2 and 4.24x at beta = 4, so successive sweeps are not independent and the permutation null understates the spread a re-pairing would really have. |z| is optimistic by roughly that factor's square root. This does not reach the seed-spread errors, which are taken across independent chains and carry the autocorrelation already.

The rows are quoted to the precision they reproduce to, which is not the same precision. The deficit is reproducible to under 1% where the claim lives and to 10% at beta = 6, so the deep rows carry fewer digits. That difference is itself the reason §7 claims onset and not severity: the deficit is a converged estimate on the sign-free rows, and on the deepest row its error does not shrink with sampling in the way a converged estimate's must (reads/expBB_deficit_convergence.py, three seeds a cell): the relative seed spread falls from **7.6% to 3.3%** where the sign is free as the sampling is doubled, and *rises* from **3.3% to 33.7%** where it is not, with the central value moving from 0.12081 to 0.20154 on the same rows. A converged estimator cannot do either.

The capability does not depend on the field distribution. Run on the same axis with the *continuous Gaussian* spin decoupling – a different measure realising the same decoupling (reads/expAL_capability_across_representations.py, mu = 0.4, 400 draws x 4 seeds):

Monotone in beta and resolved at |z| ~ 50 in both, against a permuted null that stays under 3.4, with the negative fraction identically zero on the first four rows. What is being compared is the *shape* – the two field distributions are different measures and neither deficit is a prediction of the other – and they agree to about 10% wherever the negative fraction is zero.

The last row is a limitation, and it is the first row where the sign problem appears. The Ising deficit there spans 0.09336 to 0.95636 across four seeds – a spread ten times its own lower value. The deficit is reproducible while the negative fraction is zero and stops being so at the onset, which bounds the claim to exactly the regime §7 makes it in. Reproducibility across seeds is how that boundary is located, and it is why every row here is replicated.

On the controlled axis of §5 the read gives a calibration and no trend. Along tp at fixed filling it is exact at tp = 0 and unusable after: strength is -1.0000, -0.0480, unresolved (z = -0.55), -0.2794 – not monotone, and the unresolved row is the one with the worst sign problem (0.1825). The library returns zero there rather than a value, marking the read unresolved.

A monotone quantity does appear on that axis – tightness falls 1.000, 0.848, 0.676, 0.564 – and it is **not** a property of the coupling. At the symmetric point G_dn = 1 - G_up exactly, so the centred channels satisfy B~ = -A~ and the cross-covariance is -A~' A~, whose spectrum is the single channel's own. Measured directly: the dominant share of channel A alone is 0.9986 where the coupling's tightness is 1.0000, and away from the symmetric point the coupling's value sits between the two channels' individual shares (0.770/0.963, 0.772/0.423, 0.470/0.613). tightness is inherited from the channels' internal structure, which also varies with tp. It is a property of each channel on its own, not of the coupling between them.

The scope is by axis. Along beta at fixed filling the deficit is monotone and resolved at |z| ~ 130 where the average sign is identically 1 – that is the capability, and it is what §7 claims. Along tp the read supplies an exact calibration at the symmetric point and no usable signal away from it.

beta	neg fraction	Ising deficit	Gaussian deficit	|z| (both)	permuted null
1.0	**0.00000**	0.00563 - 0.00598	0.00627 - 0.00629	52.0	<= 1.7
1.5	**0.00000**	0.00902 - 0.00958	0.00985 - 0.01058	~49	<= 2.3
2.0	**0.00000**	0.01412 - 0.01480	0.01438 - 0.01530	~48	<= 1.6
3.0	**0.00000**	0.02887 - 0.03184	0.02891 - 0.03164	~47	<= 3.4
4.0	0.005 / 0.0006	0.09336 - **0.95636**	0.05197 - 0.09512	32 / 46	<= 1.6

A read of each channel against the system's own zero (`reads/expAT_balance_at_the_systems_own_zero.py`). `Screen.register` takes a lens's own laws – `entry`, `inverse`, `energy`, `zero` – and its documentation is explicit that a side withholding them inherits defaults that are *"the right laws for a system that has no others and the wrong ones for a system that does"*. This system has its own zero: at half filling `G_up[i,i] + G_dn[i,i] = 1` configuration by configuration, so each channel balances at **1/2**, the particle-hole point. The library's default is the column mean, which is the sample's own centre and says nothing about where the system balances.

`Screen.balance` scores what each side's zero leaves against exact no-drift moments – a scaled chi-square whose degrees of freedom are the covariance spectrum's participation ratio, with no tolerance anywhere. Every row below sits at `<n> = 1.00000`, so filling is not the variable; 12 independent runs per row:

The last column is an identity rather than a measurement, and that is the point. The default zero is the column mean, so the residual it scores is the centred sample's own mean – identically zero – and the pvalue is exactly `1.00000` whatever the data, which `tests/test_the_reads_are_the_statistics_named.py` pins on random columns at four offsets and sizes. Withholding the system's law does not weaken the read, it removes it, and the separation in the previous column therefore belongs entirely to the zero.

What the read is. `balance` scores the joint residual `||r||^2` against the exact no-drift moments `E||r||^2 = tr(S)/T` and `Var||r||^2 = 2 tr(S^2)/T^2`, which fix a scaled chi-square whose degrees of freedom are `(tr S)^2 / tr(S^2)` – the **participation ratio of the covariance spectrum**. It is therefore a correlation-aware joint test, and that is what separates it from the obvious alternative: a per-column deviation-of-the-mean test broadly tracks it and **orders two rows the opposite way**. On a staggered `2x4` and a flux-`pi/2` triangular ladder over six seeds the naive test puts the staggered row higher (`z = 2.85` against `2.30`) while `balance` puts the ladder higher (median pvalue `0.280` against `0.0089`). The two channels here are strongly correlated, so the effective degrees of freedom sit far below the column count and a test assuming full rank misreads them.

How strong that separation is, measured. The `31x` between those medians is a median effect and it does **not** survive seed by seed: the ladder's per-seed pvalues run `0.0017` to `0.862` and the staggered row's `2e-15` to `0.072`, so the two populations overlap and the ordering holds on five of the six seeds rather than all six. What is asserted is therefore the reversal of ordering, not a clean split, and the gate pins the overlap. The classical instrument of the same shape is Hotelling's `T^2`; what the read supplies is the rank, derived from the spectrum rather than assumed.

Read the pvalue, not the boolean. `closed` is a per-run decision at the reader's level and fluctuates – a sign-free ring fires on 9 runs of 12. The pvalue behind it is the quantity and its median across independent runs separates with no overlap, which is the same discipline every other sampled figure here uses.

Scope. This separates **real-weight** sign problems at fixed filling. It does not see a phase problem: the route B rows, whose channels are exactly related and whose weights carry a phase, read `0.49` and `0.38`. Route B leaves the channels exactly related, and this read asks each channel against its own zero, which that relation leaves undisturbed.

Why no reweighted estimator can be a severity meter (`closed/expAS_weighted_reads_are_capped.py`). Recovering a physical expectation from an `|w|`-sampled chain requires the reweighted ratio `<O s> / <s>`, and correct importance sampling draws at `|w|`, so a configuration enters that ratio carrying only its **sign**. Kish's effective sample size of those weights is then

```
ESS = (sum w)^2 / sum w^2 = (n <sgn>)^2 / n = n <sgn>^2
```

which is the `O(1/<sgn>^2)` cost of the sign problem written as a property of the weights, exactly and for every sign pattern (§11, `Ceiling.kish_of_signs`). `carriage` returns exactly that as `effective_n` – verified at a ratio of `1.0000` across eight `(n, <sgn>)` combinations – and it is documented as *a property of the weights, not of the frames, and the ceiling on what a reweighted estimator built from them can support.*

That is the sign problem restated as a property of the weights rather than of the model: **any** estimator that reweights by the sign pays it, whatever the observable. It is a statement about reweighted means, which is what Kish's formula is about, and it is not extended here to arbitrary functionals of a signed sample.

§7's coupling read is not such an estimator, and that is why it is readable. The read is handed the two channels' Green's-function diagonals and nothing else; the sign is collected alongside and never enters it (`reads/expQQ_coupling_vs_sign.py`, and asserted in `tests/test_the_reads_are_the_statistics_named.py`). It is an unweighted statistic of the configurations the `|w|` sampler

K	sign problem	deficit	**own zero**, median p	**default zero**, median p
2x4 clean	no	0.0000	**0.34104**	1.00000
ring 6, flux `pi/4`	no	0.0000	**0.56477**	1.00000
ring 8, flux `pi/4`	no	0.0000	**0.54925**	1.00000
staggered `h = 0.2`	**yes**	0.0661	**0.02489**	1.00000
staggered `h = 0.6`	**yes**	0.0644	**0.00000**	1.00000
ring 5, flux `pi/2`	phase	0.1545	0.48993	1.00000
tri ladder, flux `pi/2`	phase	0.4792	0.37657	1.00000

visits, so it does not pay the `1/<sgn>^2` cost – which is exactly why it carries a converged value on rows where `<sgn>` is `1.00000 +- 0.00000` and has no derivative to read. What it reports is a property of the ensemble the sampler actually visits, and the reason it cannot report severity is the coarseness below, not this ceiling.

The read is an onset detector rather than a severity meter. No map from the deficit to `-ln<sgn>` is calibrated here, and on this data none can be: on the rows where the deficit is useful, `<sgn>` is identically 1 and `-ln<sgn>` is identically 0, so there is nothing there to calibrate against. §8 gives the reason this is not a matter of more data – the severity is set by the parity of a crossing count, and the deficit is a summary that has already discarded that count.

The reading is a synchronisation, and the sign of it is set by the mechanism. A read that returned `-1` wherever the model is sign-free would be indistinguishable from a read that had learned one number, so it is put to the other place this model is provably sign-free. The two places are structurally opposite. Repulsive `U` decoupled in the spin channel is sign-free at half filling because `G_dn = 1 - G_up`: the channels are exact negatives, anti-synchronised. Attractive `U` decoupled in the charge channel is sign-free at **any** filling because the field couples to `n_up + n_dn - 1`, both spins see the identical diagonal factor, and the weight is `det^2`. Same instrument, same call (`reads/expAE_both_mechanisms.py`, 2x4, `|U| = 4`, `beta = 10`, 300 configurations per seed, **six seeds per row reported as a range**):

The ranges carry the result. Every saturated row is saturated on all six seeds to four decimals, and the entire seed spread lies in the unsaturated rows – the repulsive `mu = 0.4` row runs from `-0.3532` to `0.0000`, a spread the size of the value. That asymmetry is what makes the contrast readable. The replication also sets the scale for the controls: on the non-bipartite `3x4` control one seed in six shows **no** negative weight in 300 draws, so a control row is read across seeds and never from one.

What that last column is (`reads/expBI_greens_diagonal_at_large_beta.py`). `G` here is `(I + B)^-1` in the conditioned frame of §2, and its diagonal is an occupancy only while that frame is well conditioned. `G` is not symmetric, so its diagonal is not bounded by its spectrum once the eigenvector basis is ill-conditioned. Measured on this lattice at half filling, the diagonal stays inside `[0, 1]` to `beta = 6` and leaves it after – `-0.2754` to `1.1181` at `beta = 8`, `-0.5423` to `1.2563` at `beta = 10` – so the column is a maximum over 1800 draws of a quantity that is no longer a filling. The relation the §7.2 calibration rests on is unaffected and exact throughout: `max|G_up + G_dn - 1|` runs `2.7e-15` at `beta = 2` to `6.2e-14` at `beta = 12` on the same draws. The relation *between* the channels survives where neither channel's diagonal is an occupancy, and that relation is what §7 reads.

The last column is why the +1 rows are not the evidence. On the attractive rows the two channels come out bit-identical, so the read is being handed the same array twice; `+1` is the sign of the reading and says nothing about the instrument's discrimination. What the table establishes is the **contrast**. Doping is exactly what takes the repulsive read from `-1.0000` to unresolved, and it leaves the attractive read where it was – because the attractive model stays sign-free under doping and the repulsive one does not. So the read tracks whether the channels move together, which is the property the sign depends on, rather than filling or distance from a particle-hole symmetric point.

Saturation and a zero negative fraction are **not** equivalent. At `mu = 0.8, beta = 6` the negative fraction runs `0.0000` to `0.0100` across seeds while the read runs `-0.8665` to `-0.5358`: the departure from saturation exceeds the negative fraction on every seed, including the seed where the negative fraction is exactly zero. That is §7's own result – the read departs before the average sign does – appearing again.

No filling is estimated in that table. These are uniform draws over the field rather than importance samples, so no expectation value taken from them carries information; the rows are labelled by their inputs.

The boundary of the reading, measured. Both saturated cases above relate the channels *affinely*, so the test is a mechanism whose positivity comes from somewhere else. Wu and Zhang [6] and Wei et al. [7] give one: with a spin-dependent twist the two spins become time-reversal partners, `K_dn = conj(K_up)`, and with attractive `U` in the charge channel `B_dn = conj(B_up)`, so the weight is `|det(I + B_up)|^2 >= 0`. Positivity here is *conjugate pairing*, not an affine relation. The control applies the **same** twist to both spins: time reversal is gone, the weight becomes `det^2` and acquires a phase, with ev-

U	channel	mu	negative fraction	strength	max|G_up - G_dn|
+4	spin	0.0	0.0000	**-1.0000 to -1.0000**	3.1e+03
+4	spin	0.4	0.0433 - 0.0800	-0.3532 to 0.0000	1.9e+02
+4	spin	0.8	0.0033 - 0.0133	-0.8424 to -0.6143	8.4e+00
-4	charge	0.0	0.0000	**+1.0000 to +1.0000**	0.0e+00
-4	charge	0.4	0.0000	**+1.0000 to +1.0000**	0.0e+00
-4	charge	0.8	0.0000	**+1.0000 to +1.0000**	0.0e+00

erything else identical (`reads/expAF_third_mechanism.py`, attractive `|U| = 4`, `beta = 8`, 200 draws x 5 seeds):

This falsifies "saturated if and only if sign-free" in both directions. The time-reversal rows are sign-free – the conjugation is exact, the weight's imaginary part is identically zero, the negative fraction is 0.0000 – and the read does **not** saturate: 0.06 to 0.94. The control rows carry a real sign problem, up to 37.5% negative weight, and the read is **exactly 1.0000** on every seed and every geometry.

Both follow from the same thing. Under time reversal the relation between the channels is deterministic but carries *opposite signs on two orthogonal subspaces* – `+1` on the real part, `-1` on the imaginary part – and one signed scalar averages them; read blockwise, the mechanism is fully visible and exactly saturated. In the control the channels are bit-identical, so the read saturates on a degenerate input, and that model's sign problem lives in a **phase common to both channels**, which no comparison *between* the channels can reach.

7.3 Where the comparison stops: a phase common to both channels

So the correspondence between saturation and sign-freedom is a statement about real determinantal weights, whose sign is a product of two determinant signs. That is the setting every row of §§3-5 was measured in, and it is where that correspondence holds.

The reading itself goes further. A complex weight is unreachable by a **comparison** between the two channels – for a reason given below, that its phase is common to them – and is read exactly by a **one-sided** read of the weight itself. What the two-mode diagnosis further separates is whether that phase is global, in which case it cancels in `<O> = sum(O w)/sum(w)` and costs nothing – leaving a real weight whose own sign problem `resultant` then measures – or configuration-dependent, in which case it is the problem itself and no rotation reaches it.

The Kramers reading [8] is the same arithmetic as §5's route A, with one sign changed. There `G_dn = conj(G_up)`, so the real parts are exact positives and the imaginary parts exact negatives, and the read returns

```
strength = +1 - 2r
```

with `r` again the imaginary share of the centred variance. Verified to **2e-16** across five geometries and two seeds each, `r` running from `0.025` to `0.423` and the read from `0.949` to `0.154`. The spread across seeds in the table above is `r`'s spread, not noise in the instrument.

So the three mechanisms this paper measures are one formula. Route B gives `G_dn = 1 - G_up` and `r = 0`, hence `-1` exactly; route A gives `1 - conj(G_up)` and `-1 + 2r`; Kramers gives `conj(G_up)` and `+1 - 2r`. A frame carrying imaginary weight cannot read `+-1` on a single signed alignment, and how far it falls short is that weight and nothing else.

So saturation is not necessary for sign-freedom: the Kramers rows are sign-free and read `0.59` **because** their frames are 20% imaginary.

Is the boundary intrinsic, or an artifact of reading one scalar? Two things were checked, and they answer in opposite directions (`reads/expAH_boundary_is_intrinsic.py`).

The `+1 / -1` split needed the mechanism to be known. Handing the complex frame to the read directly reproduces the concatenated values – the library reduces a complex frame through the same real embedding `iota(x) = (Re x, Im x)` – and the experiment asserts that agreement. The read's `phase` field is `0.0101` and `0.2120` on the two Kramers rows, so the alignment is not purely real and the signed real part is not the whole of it; but a phase of `0.2120` rad inflates the magnitude by `1/cos(phase) = 1.023`, taking `0.4794` to `0.490`. The unsaturated reading is not a magnitude hidden behind a sign. Asking the instrument for the directions itself, via `principal_directions` on channel A, does **not** recover the split: the per-direction couplings come out `-0.462, +0.854, +0.884` and `-0.509, --, +0.232`, not `+1` and `-1`. Those are the directions of A's own variance, not the eigendirections of the A-to-B relation. So the blockwise saturation of the previous table was recovered using knowledge of the mechanism, and a mechanism-agnostic application of this read stays unsaturated.

Why the relation reads are blind to a phase: the phase is a common mode. Placing the two determinants on the screen rather than the two Green's functions shows what is happening. On every route B row the unit-modulus determinants couple at `+1.000` with `z ~ 21` – their phases are perfectly aligned. The weight is `det_up * det_dn`, so its phase is twice a phase the two channels *share*, and it varies configuration to configuration. Two aligned quantities moving together are invisible to any comparison between them, which is what every two-sided read reported.

Lx x Ly	phi/pi	T-rev	`G_dn = conj(G_up)`	`max|Im w / Re w|`	negative fraction	strength over seeds	Re only	Im only
4x3	0.0625	yes	**0.00e+00**	**0**	**0.0000**	0.8575 to 0.9445	+1.0000	-1.0000
4x3	0.0625	no	4.49e+00	3.3e+02	0.0400	**1.0000 to 1.0000**	+1.0000	+1.0000
4x3	0.1875	yes	**0.00e+00**	**0**	**0.0000**	0.4889 to 0.7346	+1.0000	-1.0000
4x3	0.1875	no	6.84e+00	2.7e+02	0.3750	**1.0000 to 1.0000**	+1.0000	+1.0000
3x4	0.1250	yes	**0.00e+00**	**0**	**0.0000**	0.0649 to 0.7039	+1.0000	-1.0000
3x4	0.1250	no	1.58e+01	7.8e+02	0.2650	**1.0000 to 1.0000**	+1.0000	+1.0000
5x2	0.1000	yes	**0.00e+00**	**0**	**0.0000**	0.6201 to 0.8618	+1.0000	-1.0000
5x2	0.1000	no	6.35e+00	1.4e+03	0.1400	**1.0000 to 1.0000**	+1.0000	+1.0000

8. What does not carry the sign

The sign is a per-configuration fact, and a per-configuration order parameter would be worth more than an ensemble one. Three candidates were derived and tested. All three fail, and they fail for the same reason.

candidate	what it is	why it fails
`max_i |G_up[i,i] + G_dn[i,i] - 1|`	the residual of the particle-hole identity	a coarse summary of the object carrying the sign. Median 1.1x to 6.4x higher on negative-weight configurations, but positive-weight ones reach 190 while negative ones go down to 0.22
`min_k | lambda_k(M) |`	the distance to a sign flip, in the conditioned frame	a **distance is sign-blind**. Negative-weight configurations sit 2 to 8x closer to a zero in median, and the distributions overlap completely: a configuration can approach a crossing without making it, or cross and travel far past
the §4 identity residual	the exact failure of an exact relation	a statement about **magnitudes**. Median ratio 0.74 to 1.10 – no discrimination at all

The reason is structural. `det(I + B) = prod_k (1 + lambda_k)`, so the sign is the **parity** of the number of eigenvalues past the crossing – exactly, and for any product (§11, `Parity.prod_pos_iff_even_neg`).

What that rules out is narrower than it first appears. The sign is a discontinuous function of the spectrum, so no continuous function *equals* it (`sign_not_continuousAt_zero`, `no_continuous_function_is_the_sign`). A continuous function can *determine* it: the product does, which is `det_determines_sign`, and the product is continuous in the eigenvalues. Continuity alone is therefore no obstruction, and "a parity is discontinuous" settles nothing on its own.

The obstruction is coarseness, not regularity. Each of the three candidates is a summary that has already discarded the crossing *count* – a maximum residual, a minimum modulus, a magnitude comparison – and a count modulo two cannot be recovered from a quantity that never encoded it. The one cheap-looking scalar that does keep it is the determinant itself, and its cost is what the question was asked to avoid. The §4 identity *holding* forces the signs to lock; the identity *failing* only permits them to differ, and says nothing about which configurations do.

It is the same wall §9 meets from the other side, and it is why the constrained path's node is hard: what has to be predicted is a discrete global invariant, and the available cheap observables are continuous local ones.

The parity argument covers real weights only, and the sweep that shows it says more. A parity is not continuous; a *phase* is. The model carries the instrument for testing that: both `(n_up - 1/2)(n_dn - 1/2) = 1/4 - m^2/2` and `= rho^2/2 - 1/4` are exact identities on the four states of a site, so splitting `U = (1-theta)U + theta U` and decoupling each piece in its own channel gives a family in which **every theta is the same physics** – the model's own gate – while `theta = 0` is real and `theta = 1` is complex (`reads/expAI_phase_is_not_a_parity.py`, 2x4, `U = 4`, `beta = 4`, 400 prior draws x 4 seeds):

The two complex-weight columns, and why neither is a count. `|Im w| / |w|` is bounded by 1 and is a share of the weight's magnitude; a ratio to `Re w` instead would be unbounded and would diverge wherever `Re w` passes near zero, which is exactly what a weight crossing between positive and negative does. `1 - |<w/|w|>|` is the mean-phase deficit, and it is zero when the weights are real and positive whatever their spread in magnitude.

A fraction of draws carrying a phase would need a chosen angle, and is not quoted. Counting the draws whose argument is neither near `0` nor near `pi` requires a value for "near"; at `0.1` radians it undercounts, reading `0.9350` where every draw in the row carries a phase. Removing the window does not repair it – `Im w != 0` then reads `0.9925` to `1.0000` on the `mu = 0` rows, whose weights are real, because at that point it is counting floating-point dust at `1e-15`. There is no cut-free form of it at finite precision. The two columns above need no window and separate the same two populations by fourteen orders of magnitude, so the claim is made with them.

The phase is not a function of the field either, and this is asked on the whole field. §4 gives the magnitude ratio exactly from the field, `ln|det_up| - ln|det_dn| = -dtau L tr(K) + lambda sum(x)`, so the natural question for a complex weight is whether `arg(w)` has an analogous relation. If it did, the phase could be computed without the determinant and the sign problem would not be a sampling problem at all.

A null on any particular functional says only that the one functional does not carry the phase, so the question is put on the largest linear basis available (`reads/expAX_is_`

mu	theta	mean |Im w| / |w|	1 - |<w/|w|>|	neg fraction of Re w	strength over seeds
0.0	0.00	0.0e+00	**0.0000**	**0.0000**	**-1.0000 to -1.0000**
0.0	0.50	3.4e-15	**0.0000**	**0.0000**	-0.1725 to 0.0653
0.0	1.00	2.2e-14	**0.0000**	**0.0000**	**+1.0000 to +1.0000**
0.8	0.00	0.0e+00	0.0000	0.0006	-0.8956 to -0.8895
0.8	0.50	6.4e-01	0.7159	0.4050	-0.0790 to -0.0458
0.8	1.00	6.4e-01	0.6460	0.4956	**+1.0000 to +1.0000**

the_phase_a_read_of_the_field.py): the frame is every one of the L x N Ising variables per configuration, nothing selected and nothing summarised, read by carriage – whose null is the exact re-pairing of weights with frames – against the phase's two components. Every linear functional of the field is a vector in that basis, so the read subsumes them all.

	carried	against its own null
ln|det_up| - ln|det_dn| (control)	1.95 to 2.68	**+12.1 to +22.9 sigma**
cos arg w	0.83 to 1.16	-1.8 to +2.1 sigma
sin arg w	0.84 to 1.13	-1.7 to +1.4 sigma

carried is 1 when the weights carry nothing. Every phase reading sits inside the null's own observed range across six lattices, while the magnitude control – read on the *same frames*, in the same call – stands twelve to twenty-three sigma clear of it. The per-coordinate resolved counts are not the statistic and are not quoted: at far = 0.05 over 384 coordinates about 19 clear the level by chance, which is the range those counts come out in.

The phase behaves as the sign does – the magnitudes are given by a closed form and the argument carries no linear information from the field that this instrument resolves on the full basis. What this leaves open: a nonlinear function of the field is outside a linear alignment's reach, and a relation below the instrument's resolution at 400 configurations would not register.

The mu = 0 block is the result. Same Hamiltonian, same physics, weights real and positive at every theta – there is no sign problem anywhere in that block, charge channel included – and the read runs the full range from -1.0000 to +1.0000. A quantity that moves that far while both the model and its sign structure hold still is reading the decoupling, not the model. **The read is a property of a (Hamiltonian, decoupling) pair** – as is the sign problem itself. So the read is quoted about a Hamiltonian together with the decoupling used, and every claim in §§3-7 is a claim about the *spin* decoupling named there.

The doped block supplies the complex weights the parity argument does not reach: the weights are about 60% imaginary by magnitude – mean |Im w| / |w| runs 0.568 to 0.647 there against 3e-15 on the half-filled rows of the same family – so there the sign problem is a *phase* problem. The read does not reach it either: at theta = 1 it is saturated at exactly +1.0000 while 49.6% of draws have negative real part – the collision of §7 again, since theta = 1 sets lambda_s = 0 and the two channels coincide.

And in the interior of the family there is no lockstep at all. Measuring what actually delivers positivity at each theta (reads/expAJ_how_positivity_arises.py, half filling, 300 draws x 3 seeds): at theta = 0 the determinants are real and their signs agree on **1.0000** of configurations – the mechanism of §§3-4. At theta = 1 they agree on 1.0000 trivially, because lambda_s = 0 makes them the *same number*. In between, at theta = 0.25, 0.50, 0.75, the two determinants' real parts agree on **0.5167, 0.5178, 0.5289** of configurations – chance – while the weight is exactly real and positive on every draw. The lockstep is absent there and the model is sign-free anyway. That is the mechanical reason the read decays across the family: it compares the two channels, and in the interior the channels are not what carries positivity.

One caution about that experiment. The quantity arg(charge_phase) + arg(det_up) + arg(det_dn) is arg(w), so its vanishing is positivity restated rather than an independent invariant, and its *maximum* over draws summarises nothing: a maximum of pi means one draw was negative-real, not that all draws were real. The distribution above is what the claim rests on.

No reweighted expectation is quoted from that table. Fields are drawn from the prior and the effective sample size is 1.5 to 13 of 400, which supports none. Every column is a property of the prior ensemble: the negative fraction of Re w among the draws, the typical size of Im w against Re w, and the coupling of the two channels over those draws. The §7 coupling along beta is a different object, measured on importance-sampled chains, and the two are not compared here.

Where the complex weights are. Turning the decoupling toward the charge channel does not by itself produce them: at half filling the whole family is sign-free, the charge channel included, with the weight real to 1e-15 and no negative real part at any theta. Complex weights appear under doping, which is why the table carries both blocks.

9. What any read of the sign must contend with

Each result below is a structural constraint on what can be read from sampled data, and together they are why §7

reads a coupling between two channels and claims onset rather than severity.

9.1 Reads that score severity from the sampled data. Every one reduces to the average sign under a different, cheaper measure. The anchor is where the reduction can be written down and checked (`closed/expAQ_anchor_reduces_to_the_sign.py`): with `B = diag(sigma) A`, the read returns the average of `sigma` **reweighted by each row's squared length**, agreeing with it to `8e-05` to `1.2e-03` across five draws. The residual is the centring the read performs and the plain average does not; against its own definition on the centred frames the read is exact, which is arithmetic rather than a finding. So a read of this kind does not fail to see the sign – it sees a reweighted one. They fail for one reason: the sign correlates with `|D|`, since negative weights cluster near the zeros of the determinant, so any weighting that is not `|D|`-weighted mis-weights exactly the configurations that decide the answer.

9.2 A magnitude read cannot see a sign, for any exponent (`closed/expAP_phase_blindness.py`; proved for every `q` in §11, `Blindness.sum_abs_pow_blind`). With `B = diag(sigma) A` the magnitudes agree cell for cell, so `sum |A|^q` and `sum |B|^q` agree for **every** `q` – measured at `q = 0.5, 1, 2, 3` across three frames differing by row sign flips, with a worst disagreement of **exactly 0.000e+00**, not a tolerance. The singular spectrum is blind for the same reason, `diag(sigma)` being orthogonal: maximum singular-value difference `0.000e+00`. The file carries its own positive control, and it is what makes the blindness a property of magnitude reads rather than of the frames – the coupling *between* the two sides returns `+1.0000`, `0.0000` and `-1.0000` on the same three frames. That is why §7 reads a coupling and not a magnitude: the sign of a determinantal weight lives in the relation between the channels, not in the size of either.

9.2b What can and cannot see the sign, classified. Three families of read were put to the question.

family	status
reweighting by the sign, which any physical expectation needs	**capped** at `n <sgn>^2` – §7, from Kish's effective sample size
a read of the relation between the two channels	**blind** – §7, twice: identical channels, and exactly related channels
a read of one channel alone	**measured: no ordering across lattices**

The third is the one neither impossibility rules out, so it was measured (`reads/expBG_single_channel_has_no_indicator.py`, 200 draws a point, `beta = 8`, `U = 4`). Two populations, and they answer differently.

Along one lattice's flux sweep. Threading the frustrated ladder through a full period takes the sign deficit `0.07000` up to `0.46508` at `pi/2` and back to `0.07000`, so a read that carried the sign has to turn round with it. Over those eleven points the deficit tracks itself across seeds at `rho = +0.973`. `phase` returns `+0.159` at `p = 0.640` – no relation. `top_share` and `dominance` both return `-0.770` at `p = 0.006`, so they do move with the deficit on this lattice, at well under its own reproducibility; `attenuation` returns `-0.542` at `p = 0.085`.

Across lattices. That ordering does not survive the change of population. On **eighteen** lattices running from sign-free to a deficit of `0.46508` – clean `2x4` and `2x6`, open and periodic chains, a star, four flux-threaded rings, four staggered amplitudes, a doped cell, a next-nearest-hopping cell, and two odd rings and a ladder at `pi/2` – no single-channel read orders them: `phase +0.273`, `attenuation -0.060`, `top_share +0.017`, `dominance -0.017`, every one at `p >= 0.272`.

And that null has power. At `n = 18` a rank correlation of `|rho| >= 0.468` would reach `p < 0.05`; the largest observed is `0.273`. Among the **ten sign-free lattices alone** – every one at deficit exactly `0.00000` – `attenuation` spans `0.05161` to `0.79089`, a wider range than between them and any lattice here that has a sign problem. A quantity that varies more among sign-free systems than between them and a sign problem is not an indicator of one.

What survives all three is identification of an operator from a correlation sequence, which aggregates no signed configuration and reads no channel relation. That is the family §10 points at.

9.2c The ceiling of §7 is sound but not tight, and the slack is measurable (`reads/expBC_ceiling_is_not_tight.py`). Kish's formula assumes independent draws. A DQMC run is a Markov chain, so the evidence it actually carries is `n <sgn>^2 / (2 tau_int)`, and everything above uses the ceiling as an upper bound – which it remains, since the correction only makes the true figure smaller, which strengthens the argument that no weighted read is a severity meter. The ceiling bounds the evidence a run can carry; it is not the evidence a given run holds, and it is not an error bar.

Which `tau_int` matters is the standard question with the standard answer: there is no single autocorrelation time for a chain, each estimator has its own, set by its influence function. The reported quantity is the reweighted ratio `<O s> / <s>`, whose linearisation is `z_t = (O_t s_t - r s_t) / <s>` at the pooled `r`, and `z` relaxes more slowly than `s` does. The sign's own `tau` accounts for two thirds of the gap. At `N = 8`, `U = 4`, `beta = 4`, `t2 = 0.7`, `mu = 1.0` (a real sign problem at `<sgn> = 0.827`), against a reference built from the scatter of 128 independent replicas – four seeds of 32, which is the run `--seeds 31 32 33 34`. The one-seed default gives the same picture more coarsely and different digits, so the rows below are that invocation and not the bare one:

tau of the sign	1.106 +- 0.140
tau of the influence function, per diagonal	1.50 +- 0.15 to 3.18 +- 0.43
evidence overstated by the ceiling	3.0x to 6.4x, mean 4.15x
error bar implied by the ceiling	1.98x too small
corrected by the sign's tau	1.33x
corrected by the influence function's tau	0.97x (reference tau; 1.21x from one chain – see below)

The influence function's tau is quoted per observable because that is what it is: there is one per estimator, and the eight Green's-function diagonals of this lattice do not share it. They are read off the same chains and are not independent of one another, so the mean carries no 1/sqrt(8) and the range is the result. The error-bar row is a median over the eight diagonals; the two correction rows divide that median by sqrt(2 tau) at the mean tau, so they describe a typical diagonal rather than any particular one.

The slack is ordinary autocorrelation. Holding beta = 4 fixed and switching the sign problem off with t2 = 0 gives <sgn> = 1.00000 exactly – so the ceiling reads n, the whole sample – and the slack is still 3.66x, against 4.15x with the sign problem on. A factor of 1.13x between them, with the sign problem carrying the smaller part of it. On that row the sign has no variance at all, so it has no autocorrelation time to correct by, and the slack is there regardless.

beta is the other axis the slack could be attributed to, and it was scanned (--beta, one seed a point). The slack is not monotone in beta:

beta	<sgn>	2 tau_infl
2	0.99792	2.60x
3	0.95590	3.56x
4	0.83056	4.24x
5	0.65417	3.61x

<sgn> falls monotonically, by a third across the scan, and the slack does not follow it: it rises to beta = 4 and then comes back down while <sgn> keeps falling. A quantity controlled by the sign problem could not do that. What beta does do is lengthen the imaginary-time extent and slow the chain, which is ordinary, and the t2 = 0 row above is the same statement with beta held fixed instead.

None of the machinery here is new. That Kish's formula assumes independent draws is textbook, and so is the treatment of a derived quantity: Wolff's Gamma method [17] propagates a function of primary observables by projecting their fluctuations through its derivatives, pi_F = sum_alpha (dF/da_alpha) pi_alpha, which for F = <O s>/<s> is exactly the z_t above. It has done so since 2004 and ships in the standard packages. What is reported here is therefore an applied correction to *this* paper's bound – n <sgn>^2 is stated in §9 without the autocorrelation factor that standard error analysis would supply, and on this rig that factor is 4.15x – not a new way of obtaining it.

The control identifies the chain as the cause. Drawing the same sweeps i.i.d. with replacement from the pooled ensemble preserves the marginal, and so preserves <sgn> and n <sgn>^2, while removing the dependence between consecutive entries: tau_infl falls from 2.074 to 0.536 and the sign's from 1.106 to 0.463, both the independent value of one half, and the ceiling goes from 1.98x too small to 1.02x, which is tight. Permuting each chain's sweeps in time is a different control and carries no information here: it leaves each chain's mean where it was, so the scatter of chain means is invariant by construction. Residual thermalisation is excluded separately, because a chain still relaxing would inflate the scatter through the same channel as autocorrelation: the across-replica scatter of the first half of each chain and of the second differ by a median factor of 1.07 over the eight diagonals, spanning 0.93 to 1.17. A chain still thermalising would have the first half scattering more.

tau_int is read with dynamics(z).reconstruct_decay() , which rebuilds C(tau) from the operator's modal powers and eigenvalues and sums it over all lags, so no window, truncation lag or multiplier is chosen: moving the summation limit from 1000 to 8000 changes the answer by 0.0e+00. Two limits are measured. It must be reconstruct_decay, which reads the **connected** operator; rates().dominant reads the raw one, where the constant function is an eigenmode with |mu| = 1, and on a sign sequence with mean 0.83 it returns that mode instead of the decay. And a scalar frame fits a single exponential, while this chain's influence function has a slower tail, so the one-chain read understates tau by about 1.5x: it takes the error bar from 1.98x to 1.21x, not to 1.00x. Closing the rest needs the replicas. The estimator itself is not new – window-free routes are established practice, emcee fitting a second-order ARMA for this purpose [16] and Wallerberger's log-binning targeting lattice and Monte Carlo chains [15] – and each keeps a constant of its own, a model order and a binning factor respectively. The claim made here is the measured correction to this paper's own bound, not the means of computing it.

9.3 The scalar decoupling family is complete (model2d.single_site_identity, reads/expAI_phase_is_not_a_parity.py). Using n^2 = n, the interaction has exactly two quadratic forms, with the m^2 coefficient pinned at -U/2 and the rho^2 coefficient at +U/2 for any rewriting; the "shift around alpha" freedom is one-body and is absorbed into the hopping matrix. Both rewritings are checked by quadrature on a single site at every mixing (model2d.single_site_identity, 40 Gauss-Hermite nodes): over U in {2, 4, 8}, dtau in {0.0625, 0.125, 0.25} and theta in {0, 0.25, 0.5, 0.75, 1}, all four occupation states of each cell, the worst relative error is **6.5e-16**; a single cell alone reads 2.1e-16, so the figure is a property of the whole grid.

Splitting U between the spin and charge channels is therefore the whole family, and the mixing was swept

across it. Which decoupling has the worse sign problem is a question about <sgn>, and <sgn> is an expectation, so it is **summed rather than sampled** (reads/expBJ_decoupling_family_exact.py): every auxiliary field is enumerated – both of them, the real spin field and the imaginary charge field, 4^(N L) configurations – and <sgn> comes out as |sum w| / sum |w| with no sampler, no seed and no effective sample size. On a 2x1 lattice at L = 4, dtau = 0.25, U = 4, that is 65536 field pairs a row:

At half filling every theta is sign-free exactly, which is §8's statement about this family restated without a sampler. Under doping <sgn> falls monotonically from theta = 0, and no interior theta beats the pure spin decoupling by more than 1e-12: **the spin channel wins with no interior optimum**, on an exact sum rather than on draws from the prior. The SU(2) vector decoupling lies outside the family; it was derived and built, and is 2.2x worse at 2x2 and unresolvable at 4x4.

The lattice is small because exactness costs exponentially. What the enumeration establishes is the ordering across theta, which is what the claim is; it is not a statement about how severe the sign problem becomes at production size.

9.4 A constant contour shift buys nothing here (closed/expAR_contour_2d.py). Deforming the integration contour is the one route that can change the exponent rather than the prefactor. Two shift shapes were tried, neither fitted – a uniform c, the one-dimensional deformation carried over, and one staggered by sublattice, which is the model's own structure and which a uniform shift cannot see. Measured on 4x4, mu = 1.0, U = 6, beta = 6, against an undeformed mean phase of **0.4167 +- 0.0554** re-measured in the same run, every gain is below 1: the best is 0.23 and the worst 0.08, so the deformation makes the phase problem worse at every amplitude tried. The scan runs to c = 4, far past any plausible optimum, because a thimble need not lie near the real axis and a second regime at large shift would have falsified the reading; there is none. The point was selected by measuring undeformed phases first – at mu = 0 this lattice is at §5's sign-free point, where the undeformed phase is 1.0000 and any shift can only spoil it – and the undeformed value is printed beside the scan.

What that does not close. A constant shift is the crudest member of the deformation family. A thimble method deforms along a flow, and nothing here measures one.

9.5 Complex Langevin [12, 13] (closed/expCC_cl_fails.py, closed/expDD_cl_reads.py, closed/expEE_no_error_bar.py, closed/expJJ_seed_spread.py). Built and gated: the analytic drift against a central finite difference of the complexified action at 7.4e-10 in the spin channel and 1.4e-09 in the charge channel, with the wrong-convention control at 0.40 and 0.36 (tests/gate_main.py). It converges to the wrong answer here, and only in the charge channel. On the doped rows of closed/expDD_cl_reads.py the charge channel returns a density of 1.05095, 1.10220 and 1.17205 against exact values of 1.03977, 1.08089 and 1.13979 at mu = 0.3, 0.6, 1.0 – resolved at z = 64.4, 85.4 and 33.6 – while the spin channel on the same grid returns 1.03974, 1.08084 and 1.13987 against the same exact values, matching to better than 1e-4 on every row. The uncertainty on those figures is estimated from reproducibility across seeds; a within-run formula and a seed-to-seed spread differ by more than an order of magnitude on this data, and the seed spread is the one that holds.

9.6 The constrained path's trial wavefunction [9, 10] (closed/expW_trial.py, closed/expX_dial.py, closed/expKK_multidet_bias.py, closed/expLL_unfitted_trial.py, closed/expMM_self_consistent.py, closed/expNN_selection.py). The bias is set by the node and by nothing else, and it moves by a factor of 24 with the trial at fixed coupling – across the staggered-field family at U = 8 it runs +0.05246 at h = 0 to +1.25917 at h = 4. The computable criterion points the wrong way: argmin <Psi_T|H|Psi_T> is *anti*-correlated with the bias over a nine-candidate pool (rho = -0.80, p = 0.0096). Fitted to the exact ground state, four to six non-orthogonal determinants cut the bias by 1.7x to 10.4x depending on the coupling, so the prize is real – and **no answer-free construction beat the plain free determinant**, every candidate coming in 5 to 15x worse. What blocks the route is not generating determinants but that the only objective which selects them correctly is the one that needs the answer.

Two things make that band a band rather than a figure. The ceiling is not monotone in k. Relative to the single-determinant walk, a run of closed/expKK_multidet_bias.py gives 0.970, 0.379, 0.044 at U = 4 for k = 2, 4, 6, and 1.165, 0.493, 0.079 at U = 8: the second determinant helps at weak coupling and *hurts* at strong. Only the U = 12 reversal clears its own error bars on that run – 3.2 standard errors from the quoted biases, against 1.0 at U = 8 and 0.1 at U = 4. A second run of the same script gives 0.777, 0.398, 0.072 at U = 4, so the direction survives a re-run and the digits do not.

The fit is why. It does not reproduce across processes (closed/expBF_multidet_reproducibility.py): three refits inside one process agree to all seventeen significant figures, a spread of exactly 0.0e+00, while four separate processes return four distinct values spanning of order 1e-7 – enough to move the bias by more than the walker seed spread, so the largest reduction reads 10.43x on one full run and 9.00x on another. **The error bar recorded beside each entry understates it**, because it varies walker seeds with the fit held fixed and has no term for the fit. The low end is stable across runs (1.74x and 1.73x); the high end is quoted as a band.

10. Outlook

The reframing changes what a positivity-restoring representation has to achieve.

theta	<sgn> at mu = 0	<sgn> at mu = 0.8	mean-phase deficit	negative fraction of Re w
0.00	**1.00000000**	**1.00000000**	0.00000000	0.00000000
0.25	**1.00000000**	0.98315847	0.02467001	0.00000000
0.50	**1.00000000**	0.96202358	0.06589094	0.00122070
0.75	**1.00000000**	0.93210857	0.16293791	0.04254150
1.00	**1.00000000**	0.85036742	0.38931226	0.15625000

Not avoidance – synchrony. A natural reading of a sign-free model is that its configurations stay away from the boundary where a determinant vanishes. That is measurably false here, and the way it fails is the evidence. At `beta = 10` and half filling the distance distribution has a bulk that does not move and a tail that reaches the boundary: the median holds at **0.50 to 0.55** across every sample size and seed, while the 1st percentile sits **25 to 77 times below it**, at `0.007` to `0.020`. The smallest distance observed has no stable value at all – it ranges over `0.0002` to `0.012` across seeds at fixed sample size, and falls further as draws are added – which is itself the statement that no distance is being respected. Throughout, the negative fraction is exactly **0.0000**.

So configurations reach arbitrarily close to the boundary, and it is a tail doing it rather than the bulk drifting inward. The individual channels cross on around 11% of configurations. What the symmetry supplies is not distance but **synchrony** – the two channels cross together.

The criterion says what a construction has to achieve. §5 turns the requirement into a decidable property of the one-body matrix: a diagonal unitary `S` with either `S K S^-1 = -K` or `S K S^-1 = -conj(K)`. It is constructive on the hopping graph – it names the flux that opens a route where frustration closes every flux-free one – but only route A leaves the weights real, so on a graph with an odd cycle the identity can be restored and the positivity cannot.

It is also a **closed door on doping**, in one line. Every diagonal matrix commutes with every diagonal conjugation, so `S K S^-1` leaves `diag(K)` untouched; both routes then demand `K_ii = -K_ii` – and `K_ii` is real because `K` is Hermitian – so `diag(K) = 0` is forced. A chemical potential can therefore never be accommodated, by any flux, on any graph, at any size. Measured over 84 flux values – rings of 5, 6, 7 and 8 sites at 21 fluxes each, a full period on every graph – the best residual of all 84 is `1.776e-14` at `mu = 0`, and moving the diagonal takes it to `8.926` at `mu = 0.2` and `26.63` at `mu = 0.6` (`reads/expBK_doping_closes_every_flux.py`). Flux is the only freedom a diagonal-unitary conjugation has on a cycle, so sweeping it over a period sweeps everything route B could use. The measurement samples; the statement quantifies, so it is also proved – §11, `Doping.doping_closes_both_routes`.

So a construction seeking positivity away from half filling cannot get there by choosing a lattice, a flux or a size: the obstruction is a diagonal term, and no conjugation of this kind removes one. It does not need to push the spectrum away from anything either. It needs to reproduce a **pairing** between two channels whose magnitudes are unrelated over 8 to 16 orders and whose crossings coincide exactly. That is what the algebraic positivity results – Majorana, Kramers, split-orthogonal [7, 8] – achieve when they achieve it, by making the weight a squared modulus so the question cannot arise; it is also what a meron-cluster decomposition [11] achieves, by making the cancelling configurations cancel exactly rather than estimating their residue; and it is why a perturbative or reweighting fix has nothing to expand in, since there is no small residual, only a discrete coincidence that either holds or does not.

What a read of this problem can carry, classified. Three families were put to the question. Each entry is established in the section named, and the first two are proofs rather than measurements – machine-checked ones, in the sense of §11.

The ceiling is the sign problem restated: correct importance sampling draws at `|w|`, so a configuration enters any weighted average carrying only its sign, and Kish's effective sample size of those weights is exactly `n <sgn>^2`. Because that number is a property of the **weights**, it bounds every estimator that reweights by them. §7's coupling read does not reweight – it is an unweighted statistic of the configurations the `|w|` chain visits – which is why it is readable where `<sgn>` is not; what stops it reporting severity is §8's coarseness – the read has already summed away the crossing count the severity is set by.

The blindness is two distinct statements. Where the two channels coincide as data, no comparison between them remains. Where they are distinct and **exactly related** – §5's route B, at `5e-15` – the comparison is well posed and still carries nothing, because the relation holds whether the weight is positive or spread across the circle.

What survives is the last row, for a structural reason: identifying an operator from a correlation sequence aggregates no signed configuration and compares no channels, so neither the ceiling nor the blindness applies to it. It is a place to look rather than a result – what it costs to identify such an operator from data a sign problem permits is not settled here.

One attempt on that route is recorded, and it failed (`reads/expAY_order_from_the_instrument.py`). The prize is concrete: a correlation sequence `C(tau) = sum_i c_i lambda_i^tau` is measured at short `tau`, where the sign is still mild and sampling is cheap, and the eigenvalues it identifies fix the operator that gives the long-

family	status	where	machine-checked
reweight by the sign	**capped** at `n <sgn>^2`	§7	`Ceiling.kish_of_signs`
read the relation between the two channels	**blind**, two ways	§7	`Blindness.sum_abs_pow_blind` for the magnitude half
read one channel alone	**no indicator**, measured	§9.2b	–
identify an operator from a correlation sequence	not excluded by any of the above	–	–

time behaviour one would otherwise pay `1/<sgn>^2` to sample. The bottleneck is **model order** – how many `lambda_i` to keep – and every standard answer is a chosen number: an information criterion picks a penalty, a singular-value cut picks a level, a stability window picks a width. Under this paper's rules none is admissible.

`spectral_optics` reports `resolved_modes`, a count against a floor derived from the data rather than supplied, which is model-order selection performed by the read. It does not work here, and the reason is structural: on the Hankel embedding of a sum of decaying exponentials it returns **1 on all four test cases** – true orders 2 and 3, well separated and close, at each of four noise levels from none to `1e-2`, five seeds apiece – while the same matrix has exact numerical rank 2 and 3 with singular values an order of magnitude apart (5.42 and 0.46 at order 2). The leading mode carries a `top_share` of 0.994 to 0.996 there: the modes are real and wildly unequal, and a floor that separates signal from a noise sea is asking a different question from "how many modes are there".

So the row stays open, with one route into it closed by measurement.

What the measurement supplies. The order parameter of §7 gives an exactly calibrated, continuously varying measure of distance from the protecting symmetry, reproducible to under 1% across seeds in a regime where the average sign is identically 1 and has no derivative. It does not forecast severity: §8 gives the reason no summary of this kind reaches the severity – the severity is set by the parity of a crossing count, and a count modulo two cannot be recovered from a quantity that never encoded it.

The open question this leaves. Whether the synchrony can be restored by construction, rather than measured. Every route in §9 that attempted it worked on the field, the contour, or the trial wavefunction, and the identity of §4 says the object to work on is neither: it is the relation between the two channels' determinants. Nothing in this work has attempted to build a representation in which that relation is imposed rather than inherited from a symmetry.

11. What is machine-checked

Several of this paper's statements are algebra rather than measurement, and a few are **universally quantified** – "at any size", "on any graph", "for every exponent" – where what supports them in the text is a sweep. A sweep over 84 flux values reports that no flux it tried opened a route; the claim being made is that none exists.

Those statements are machine-checked in Lean 4 / Mathlib, in `research/lean/` – 80 theorems and lemmas across twelve modules. The development declares no axiom of its own and contains no `sorry`. Every theorem the table below names – 48 of the 80 – carries an explicit `#print axioms`, and each elaborates against Lean's three foundational axioms alone: `propext`, `Classical.choice`, `Quot.sound`. The other 32 are the supporting lemmas those theorems are built from, elaborated by the same build. The footprint is emitted while a declaration elaborates, so it is a property of the proof and not a claim made about it.

What is not in Lean. Every number read from a running sampler. The lockstep at `beta = 12`, the `-1.0000` calibration, the measured deficits, the 15-of-15 agreement between the oracle and the read – these are properties of this model measured on this rig, and formalising them would mean formalising the rig.

What the development covers, beyond the model's own algebra, is the read: not what it returned here, but what it is guaranteed to return. §6 rests on the alignment being invariant to an offset and a scale, and on saturation being equivalent to an exact affine relation, with the deficit's `[0, 2]` range proved alongside them. Those were prose, and they are `Alignment.lean` now. With them the agreement on 15 lattices becomes a check that this rig computes the criterion correctly, rather than the evidence that the criterion works.

The theorems are about the read that is actually called. A proof about a normalised alignment and an implementation computing something slightly different would be two correct halves and one wrong whole, and neither half can reveal that on its own. The three guarantees are therefore put to `coupling.strength` itself and hold to `1e-12`: invariance to an offset and a positive scale, saturation exactly on affine data, and the `[0, 2]` bound. Each is checked against a case that must fail it – an uncentred cosine breaks the invariance – so the agreement is shown able to discriminate (`tests/test_the_read_has_the_proved_properties.py`).

The development covers the algebraic spine and what the instrument guarantees. The result is the measurement.

Two claims moved from measurement to proof.

The first is §10's **closed door on doping**. The text argues it in one line and then measures 84 flux values across four graphs, reporting a best residual of `8.926` at `mu = 0.2` and `26.63` at `mu = 0.6` against `1.776e-14` at `mu = 0`. `Doping.doping_closes_both_routes` says that for every finite lattice, every Hermitian `K` with zero diagonal, every non-zero `mu` and every diagonal unitary `S`, neither route is open. The mechanism is `conj_diagonal_eq` – a diagonal conjugation leaves `K_ii` exactly where it was – and route B needs one further step, that Hermiticity makes `K_ii` real, so `K_ii = -conj(K_ii)` forces zero rather than merely forcing it imaginary.

The second is §9.2's **blindness at every exponent**. The text measures `q = 0.5, 1, 2, 3` with a worst disagreement of exactly `0.000e+00`. `Blindness.sum_abs_pow_blind` covers every real `q` at once, so the cube is provably as blind as the square. Its companion `gram_blind` says the same of the singular spectrum, for every frame rather than for the three measured. `coupling_is_not_blind` fixes the scope: the two-sided read does see the flip, so the blindness is a property of magnitude reads.

claim (section)	Lean	status
a negative weight is a channel disagreement, and the two sets coincide (§3)	`Weight.lean`	proved – `weight_neg_iff`, `negative_set_eq_disagreement_set`, and the counted form the last table column reports as "exact"
`det(I + B) = det(B) det(I + B^-1)` for invertible `B` (§4)	`Identity.lean`	proved – `det_one_add_eq`, the step that carries the diagonal factor into the ratio of the full determinants
the log-determinant identity `ln|det(I+B_up)| - ln|det(I+B_dn)| = -dtau * L * tr(K) + lambda * sum(x)` (§4)	`Identity.lean`	proved – `log_det_identity`, with §5's pairing entering as a named hypothesis rather than as a lemma, so what the identity requires is visible in the statement
the difference cannot be constant, so no `det_up = c det_dn` exists (§4)	`Identity.lean`	proved – `difference_not_constant`, §4's opening argument
the two routes, entry by entry; they coincide for real `K` (§5)	`Criterion.lean`	proved – `routeA_iff`, `routeB_iff`, `routeA_iff_routeB_of_real`
the criterion IS a 2-colouring of the support graph with a zero diagonal (§5)	`Criterion.lean`	proved – `signed_criterion`
the spectral criterion is strictly weaker (§5)	`Criterion.lean`	proved – `spectral_similarity_is_weaker` exhibits a similarity to `-K` on a matrix with a non-zero diagonal, which no diagonal conjugation can have
a proper signing alternates along every walk; an ODD cycle admits none (§5)	`Cycles.lean`	proved – `signing_alternates`, `no_proper_signing_of_odd_closed_walk`; the tree case is the same lemma with no odd closed walk to obstruct it
a diagonal term closes both routes, at any size (§5, §10)	`Doping.lean`	proved – `conj_diagonal_eq`, `routeA_forces_zero_diagonal`, `routeB_forces_zero_diagonal`, `doping_closes_both_routes`; the staggered-potential case is the same theorem with a site-dependent diagonal
`ESS = (sum w)^2 / sum w^2 = n <sgn> ^2` exactly, and it is a ceiling (§7, §9)	`Ceiling.lean`	proved – `kish_of_signs` for every `n` and every sign pattern; `kish_le_card` (Cauchy-Schwarz) and `kish_eq_card_of_sign_free` say it is saturated exactly where there is no sign problem
a magnitude read is blind at every exponent, and so is the singular spectrum (§9.2)	`Blindness.lean`	proved – `sum_abs_pow_blind` for every real `q`; `gram_blind`; `coupling_is_not_blind` is the positive control that fixes the scope
the coupling read IS the `|x|^2`-weighted average sign (§9.1)	`Coupling.lean`	proved – `frobenius_inner_eq_weighted_signs`, `coupling_is_weighted_mean_sign`; `uniform_rows_give_plain_mean_sign` collapses it onto `p_neg` when the weighting disappears
the scalar decoupling family is complete, and its two coefficients are forced (§9.3)	`Decoupling.lean`	proved – `mixing_is_exact_for_every_theta` (every `theta` is exact, so a sweep over `theta` is the whole family), `spin_coefficients_are_forced`
the sign IS the parity of the crossing count, for any product (§8)	`Parity.lean`	proved – `prod_pos_iff_even_neg` and `prod_neg_iff_odd_neg`, by induction on the factors
what the parity argument rules out, and what it does not (§8)	`Parity.lean`	proved – `sign_not_continuousAt_zero` and `no_continuous_function_is_the_sign` (no continuous function equals the sign) against `det_determines_sign` (a continuous one determines it). The obstruction is coarseness rather than regularity

claim (section)	Lean	status
the alignment read is invariant to an offset and a positive scale (§6)	`Alignment.lean`	proved – `strength_affine_invariant`, with `centre_add_const` and `centre_smul` underneath it. This is what lets the criterion run on output with no `K`: §4's identity carries an offset `-dtau L tr(K)` and a slope `lambda`, and neither survives to the read
saturation at 1 is exactly an affine relation between the two columns (§6)	`Alignment.lean`	proved – `abs_strength_eq_one_iff`, an equivalence via Cauchy-Schwarz's equality case, with `abs_strength_eq_one_of_affine` as the direction the criterion runs in and `no_affine_relation_of_abs_strength_ne_one` as the one the build check uses
the read lies in `[-1, 1]` and the deficit in `[0, 2]` (§7.2)	`Alignment.lean`	proved – `abs_strength_le_one` (Cauchy-Schwarz, no hypothesis at all) and `deficit_mem_Icc`
§7.2's calibration is forced, not measured: `G_dn = 1 - G_up` makes the read exactly `-1`	`Alignment.lean`	proved – `strength_eq_neg_one_of_reflected` and `deficit_eq_zero_of_reflected`. A reflected column centres to the negative of the original, so the read has no other value available to it; the paper's `-1.0000` is an entailment, not a coincidence
the criterion is not vacuous	`Alignment.lean`	proved – `centre_sample_ne_zero` exhibits a column the read is defined on, and `not_affinely_related` exhibits two columns that are not affinely related, so neither side of the equivalence is empty
a sign is the rank-one case of a phase; de-rotation is exact and preserves `|<w>|` (§7.1)	`RankOne.lean`	proved – `derotation_is_exact`, `derotation_preserves_mean_modulus`, `rank_one_phase_is_two_valued`; `real_part_estimator_is_not_rotation_invariant` is the mechanism behind the `0.93500 -> 0.25011` fall

What the Lean development does not close. It does not prove the criterion is necessary for positivity, only that it decides the identity; §5 is explicit that the identity delivers positivity along route A alone, and that scope is measured rather than derived. It does not touch §7's reading, §2's conditioning, or any of §9's closed routes beyond the two algebraic ones above. And it says nothing about the open question of §10, which is a question about constructions that do not yet exist.

References

Every entry below was checked against the publisher's or arXiv's own record, and each is annotated with where this paper leans on it.

The method.

[1] R. Blankenbecler, D. J. Scalapino and R. L. Sugar, *Monte Carlo calculations of coupled boson-fermion systems. I*, Phys. Rev. D **24**, 2278-2286 (1981). doi:10.1103/PhysRevD.24.2278 — the determinantal formulation this paper measures in.

[2] J. E. Hirsch, *Discrete Hubbard-Stratonovich transformation for fermion lattice models*, Phys. Rev. B **28**, 4059(R) (1983). doi:10.1103/PhysRevB.28.4059 — the decoupling whose own constant `lambda = arccosh(exp(dtau U / 2))` is the coefficient in §4's identity, and whose two channels (spin and charge) are the two mechanisms compared in §7.

[3] Z. Bai, C.-R. Lee, R.-C. Li and S. Xu, *Stable solutions of linear systems involving long chain of matrix multiplications*, Linear Algebra Appl. **435**, 659-673 (2011) — the conditioning problem of §2, and why `I + B` has to be inverted through a stratified factorisation rather than formed.

The sign problem.

[4] E. Y. Loh Jr., J. E. Gubernatis, R. T. Scalettar, S. R. White, D. J. Scalapino and R. L. Sugar, *Sign problem in the numerical simulation of many-electron systems*, Phys. Rev. B **41**, 9301 (1990). doi:10.1103/PhysRevB.41.9301 — the exponential decay of the average sign, and the paper's own statement of the exception this one is about: the decay holds *"unless the measure is forced to be positive by an explicit symmetry."*

[5] M. Troyer and U.-J. Wiese, *Computational complexity and fundamental limitations to fermionic quantum Monte Carlo simulations*, Phys. Rev. Lett. **94**, 170201 (2005). doi:10.1103/PhysRevLett.94.170201; arXiv:cond-mat/0408370 — NP-hardness of the general problem, which is why §10 claims a reading and not a solution.

[19] R. Mondaini, S. Tarat and R. T. Scalettar, *Quantum critical points and the sign problem*, Science **375**, 418-424 (2022). doi:10.1126/science.abg9299; arXiv:2108.08974 — the average sign read as a diagnostic rather than only an obstacle, linked quantitatively to quantum critical behaviour. Cited in §1 to mark the boundary of this paper's claim: where `<sgn>` varies it is informative, and the reading offered here is for the regime where it does not vary at all.

Where positivity comes from.

[6] C. Wu and S.-C. Zhang, *Sufficient condition for absence of the sign problem in the fermionic quantum Monte Carlo algorithm*, Phys. Rev. B **71**, 155115 (2005). doi:10.1103/PhysRevB.71.155115; arXiv:cond-mat/0407272 — positivity from the eigenvalues of the fermion matrix pairing into complex conjugates, tied to time-reversal symmetry.

[7] Z. C. Wei, C. Wu, Y. Li, S. Zhang and T. Xiang, *Majorana positivity and the fermion sign problem of quantum Monte Carlo simulations*, Phys. Rev. Lett. **116**, 250601 (2016). doi:10.1103/PhysRevLett.116.250601; arXiv:1601.01994 — a unified account of the

known sign-free models, and the reason §5's scope is stated as a measured boundary rather than a general one: sign-free models exist *with repulsive interactions and without particle-hole symmetry*, so the symmetric point this paper reads is one place positivity comes from and not the only one.

[8] Z.-X. Li, Y.-F. Jiang and H. Yao, *Majorana-time-reversal symmetries: a fundamental principle for sign-problem-free quantum Monte Carlo simulations*, Phys. Rev. Lett. **117**, 267002 (2016). doi:10.1103/PhysRevLett.117.267002; arXiv:1601.05780 — the classification into Majorana and Kramers classes.

[18] I. Hen, *Determining quantum Monte Carlo simulability with geometric phases*, Phys. Rev. Research **3**, 023080 (2021). doi:10.1103/PhysRevResearch.3.023080 — a QMC simulation is sign-problem-free exactly when the total phases along the chordless cycles of the weighted graph whose adjacency matrix is the Hamiltonian vanish. Cited in §5 as the nearest statement of the same shape as route B's cycle condition, and to mark the two differences: the graph there is built from the Hamiltonian and the conclusion is positivity, where route B's graph is the support graph of the one-body matrix and the conclusion is §4's identity, which §5 shows is a weaker thing to obtain.

Routes measured and closed (§9).

[9] S. Zhang, J. Carlson and J. E. Gubernatis, *Constrained path Monte Carlo method for fermion ground states*, Phys. Rev. B **55**, 7464-7477 (1997). doi:10.1103/PhysRevB.55.7464 — the constrained-path method benchmarked in §9.

[10] S. Zhang and H. Krakauer, *Quantum Monte Carlo method using phase-free random walks with Slater determinants*, Phys. Rev. Lett. **90**, 136401 (2003). doi:10.1103/PhysRevLett.90.136401; arXiv:cond-mat/0208340 — the phase-free extension.

[11] S. Chandrasekharan and U.-J. Wiese, *Meron-cluster solution of fermion sign problems*, Phys. Rev. Lett. **83**, 3116-3119 (1999). doi:10.1103/PhysRevLett.83.3116 — a solution that works by making the cancelling configurations cancel exactly, rather than by estimating their residue.

[12] G. Parisi, *On complex probabilities*, Phys. Lett. B **131**, 393-395 (1983) — stochastic quantisation for complex actions.

[13] G. Aarts, *Can stochastic quantization evade the sign problem? The relativistic Bose gas at finite chemical potential*, Phys. Rev. Lett. **102**, 131601 (2009). doi:10.1103/PhysRevLett.102.131601 — the complex-Langevin route attempted in §9.

The instrument.

[E] Entroptics, version 0.2.10 (released 2026-11-08), version DOI 10.5281/zenodo.24063501; the concept DOI 10.5281/zenodo.21273400 resolves to the latest release. Every figure here was read through version 0.2.10. Every read is reached through `research/code/entroptics_adapter.py`, which names each one after the question this paper asks of it. The counts below are how many of the 54 cited experiments call each. `denoise` is listed in the note below rather than here, because no figure in this paper is read with it.

The coupling read is standardised against its own exact re-pairing null – exact by construction rather than asymptotic, which is what lets §7 quote a sigma with no constant supplied. `carriage.effective_n` is Kish's effective sample size and gives §9's ceiling; `reconstruct_decay` rebuilds the autocorrelation from the one-step operator's spectrum and supplies the `tau_int` that shows the same ceiling is not tight. Where a Tracy-Widom comparison is made, the external reference is [14].

One read is named above and used by no figure here. `denoise` (`Aperture.extract`) was scored as an estimator of a mean in `reads/expRR_natural_noise.py` and is the wrong tool for this problem: it splits a frame losslessly into its resolved modes and a residual, and the resolved modes carry no mean, so the mean it returns is the raw block mean to round-off; the sign problem's difficulty is entirely in that mean.

[14] M. Chiani, *Distribution of the largest eigenvalue for real Wishart and Gaussian random matrices and a simple approximation for the Tracy-Widom distribution*, J. Multivariate Anal. **129**, 69-81 (2014); arXiv:1209.3394.

[15] M. Wallerberger, *Efficient estimation of autocorrelation spectra*, arXiv:1810.05079 (2018) — log-binning of the autocorrelation function, aimed at lattice and Monte Carlo chains. Cited in §9 for the point that window-free estimation of `tau_int` is established practice; its binning factor is the constant it keeps.

[16] D. Foreman-Mackey et al., *Autocorrelation time estimation*, `emcee` documentation, https://emcee.readthedocs.io/en/stable/tutorials/autocorr/. Cited in §9 for the same point: it fits a second-order ARMA (via `celerite`) to the chain and sums the model's autocorrelation analytically, so no window is chosen – the model order is. Consulted for the method it describes, not for a numerical result.

[17] U. Wolff (ALPHA Collaboration), *Monte Carlo errors with less errors*, Comput. Phys. Commun. **156**, 143-153 (2004); erratum **176**, 383 (2007); arXiv:hep-lat/0306017 — the `Gamma` method. Cited in §9 because it already does what that section's correction needs: the autocorrelation time of a derived observable is computed by projecting the primary fluctuations through the function's derivatives, `pi_F = sum_alpha (dF/da_alpha) pi_alpha`, which is the linearisation used there. Consulted through its documented formulation in the `pyobs` implementation (https://mbruno46.github.io/pyobs/intro/gamma.html) rather than the printed paper, and cited for that formulation only.

The identity of §4 and the lockstep of §3 are derived and verified here, and no source is cited for them. Every entry above was read before being cited.

adapter	library read	§	calls
`channel_alignment`	`reads.coupling`	7.2	18
`weight_cloud`	`reads.concentration`	7.1	2
`cloud_axes`	`reads.principal_directions`	7.1	2
`evidence_ceiling`, `frame_carriage`	`carriage`	9, 8	1 each
`autocorrelation_time`	`dynamics(...).reconstruct_decay`	9.2c	1
`balance_at_own_zero`	`Screen().balance`	7	1
`single_channel_optics`	`reads.spectral_optics`	9.2b, 10	2
`dynamics`	`dynamics(...)`, for `rates` and `forgetting`	9.2c, 9.5	2